



Review Article

A review of chemical signaling mechanisms underlying quorum sensing and its inhibition in *Staphylococcus aureus*

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ABSTRACT

Staphylococcus aureus is a significant bacterium responsible for multiple infections and is a primary cause of fatalities among patients in hospital environments. The advent of pathogenic bacteria such as methicillin-resistant *S. aureus* revealed the shortcomings of employing antibiotics to treat bacterial infectious diseases. Quorum sensing enhances *S. aureus*'s survivability through signaling processes.

Targeting the key components of quorum sensing has drawn much interest nowadays as a promising strategy for combating infections caused by bacteria. Concentrating on the accessory gene regulator quorum-sensing mechanism is the most commonly suggested anti-virulence approach for *S. aureus*. Quorum quenching is a common strategy for controlling illnesses triggered by microorganisms since it reduces the pathogenicity of bacteria and improves bacterial biofilm susceptibility to antibiotics, thus providing an intriguing prospect for drug discovery.

Quorum sensing inhibition reduces selective stresses and constrains the emergence of antibiotic resistance while limiting bacterial pathogenicity. This review examines the quorum sensing mechanisms involved in *S. aureus*, quorum sensing targets and gene regulation, environmental factors affecting quorum sensing, quorum sensing inhibition, natural products as quorum sensing inhibitory agents and novel therapeutical strategies to target quorum sensing in *S. aureus* as drug developing technique to augment conventional antibiotic approaches.

1. Introduction

Staphylococcus aureus is a prominent worldwide infectious agent that leads to a significant and complicated in assessing the multitude of moderate serious dermatological infections, as well as chronic and occasionally life-threatening illnesses of the bloodstream, bones, and respiratory system [1]. *S. aureus* is a gram + ve bacterium that triggers various healthcare-associated illnesses in communities and hospital

environments [2]. *S. aureus* is a sphere-shaped, immobile bacterium notable for a wide range of pathogenicity characteristics and exceptional capability to develop resistance to many different kinds of drugs [3]. Even though there is a current decrease in *S. aureus* hospital-related diseases, *S. aureus* continues to be one of the most serious dangers to weakened immune system patients, those who underwent surgery, or people with any implanted healthcare device [4]. In the United States, the mortality rate owing to *S. aureus* infection by itself has risen above

Abbreviations: MRSA, Methicillin-Resistant *Staphylococcus aureus*; MABs, Monoclonal antibodies; QS, Quorum sensing; QQ, Quorum quenching; RIP, RNIII-hindering peptides; Agr, Accessory gene regulator; AIP, Autoinducing peptide; LuxS, S-ribosehomocysteine lyase (LuxS); TSS, Toxic shock syndrome; SEs, Staphylococcal enterotoxins; TSST-1, TSS toxin-1; PSMs, Phenol soluble modulins; ROS, Reactive oxygen species; EPIs, Efflux pump inhibitors; APDT, Antimicrobial photodynamic therapy; PS, Photosensitizers.

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20,000 fatalities per year in recent times [5]. The increasing prevalence of multi-drug resistant bacteria, like MRSA (Methicillin-Resistant *S. aureus*), makes therapy challenging. *S. aureus* can exhibit severe infectious characteristics (MRSA) and is a serious worldwide health concern [6].

In addition, the death rate, morbidity, and expenditures associated with *S. aureus* sepsis are greatly exacerbated by increasing resistance to certain most substantial anti-staphylococcal medicines available in the market, such as penicillins and methicillin [7]. *S. aureus* is prevalent in natural settings and typical human ecosystems, and it can be identified on the epidermis and mucous membranes, generally the nasal region of most healthy individuals [8]. *S. aureus* rarely results in illness on healthful skin, but if these microbes can reach the tissues inside the body or blood circulation, they may trigger several potentially harmful ailments [9].

S. aureus is the main cause of many infections affecting humans, which include viable bacteria in the blood [10], inflammation of the endocardium [11], skin and soft tissue septicity [12], acute or chronic infection to the bones [13], painful infections in the joints [14], implanted device infections [15], lung infections [16], inflammation of digestive system [17], inflammation of brain and spinal cord membranes [18], toxic shock syndrome [19], and infections in the urinary system [20]. *S. aureus* is a frequently occurring zoonotic infectious agent that can propagate among humans and animals and lead to viral infections [21].

The WHO classifies MRSA as a very dangerous pathogen, with hospital MRSA, community MRSA, and livestock MRSA all posing significant public health and safety risks. The inappropriate application of antimicrobial medicines in the livestock sector has greatly aided the spread of MRSA in animals [22]. Since resistance to antimicrobial medications is increasing, there is much concern about developing antibiotic replacements. For Gram + ve bacterium like *S. aureus*, the scenario might not be that serious compared to certain newly discovered pan-immune Gram-ve bacteria [23]. For treating *S. aureus* bacterial infections, non-bactericidal medications with curative potential or in tandem with anti-staphylococcal medications could be extremely beneficial. It is particularly relevant for infections brought on by MRSA, as methicillin is significantly more effective than the drugs where MRSA is still sensitive [24].

Initial clinical investigation has focused on anti-virulent strategies to develop antibiotic substitutes for treating *S. aureus* infectious diseases [25]. Several attempts are being made to employ monoclonal antibodies (MABs) to counteract the effects of *S. aureus* several leukotoxins and alpha-toxin; however, the findings remain unsatisfactory, or the clinical studies were discontinued, most likely as an outcome of the negative preliminary findings [26,27]. In *S. aureus*, immunity to other drugs and tolerance to conventional beta-lactam antibiotics (penicillin and variants) susceptible to beta-lactamase is almost ubiquitous [28].

Certain medications target particular *S. aureus* virulent factors, like *S. aureus* pigment and staphyloxanthine [29,30]. This is because almost all *S. aureus* virulence factors, including toxins, are regulated by the

Quorum sensing (QS) system Agr (accessory gene regulator) [31]. Much effort is endured in developing quorum-quenching techniques, i.e., methods to hinder the quorum-sensing regulation of *S. aureus* [32]. The *agr* mechanism is present in a wide range of microbial species that are significant in the healthcare and industrial sectors, involving species of the *Staphylococcus*, *Clostridium*, and *Listeria* genera [33]. When *S. aureus* reaches a certain cell population density, the *agr* mechanism is triggered, and the bacteria then enhance an extensive arsenal of secretory toxins and extracellular enzymes that aid in evading the immune systems of the host, stimulating tissue encroachment and acquiring more carbon and energy resources from the host species [34].

S. aureus employs *agr* to trigger the upregulating of virulence factors and *agr* operon when a certain level of cell density is achieved, as shown in Fig. 1A. After membrane transport and additional processes, the matured peptide is diffused outside the cell setting as an autoinducing peptide (AIP) signal, which functions as the chemical facilitator of *agr*-type QS. *S. aureus* upregulates virulence indicators and the *agr* operon when cell density reaches a certain threshold. AIP, a class of tiny signaling peptides, mediates this QS mechanism [35].

The homologous AIP structures for the four *S. aureus* *agr* subgroups (I–IV) as shown in Fig. 1B. The alphabets represent single-letter codes for amino acids for all AIP subgroups. Inhibiting the *agr* mechanism in *S. aureus* is recommended as an anti-virulence technique for various therapeutic and industrial endeavors. Hence, this review presents an exhaustive overview of various molecular processes underpinning the QS mechanism in *S. aureus*. This review sheds light on the QS in *S. aureus* and how to inhibit it by targeting the operon's regulating genes. This review, moreover, explores QS targets and gene regulation, environmental parameters that influence QS in *S. aureus*, QS inhibition, Innovative natural-derived QS inhibitors to combat *S. aureus* infections, and novel therapeutical approaches to target QS in *S. aureus*. These tactics are very helpful for reinventing to enactor medications and hindering the appearance of resistance in *S. aureus* towards drugs, which will benefit humanity and healthcare environments.

1.1. Sources of developing *S. aureus* infections

S. aureus infectious diseases are mainly caused by unidentified colony formation or, more occasionally and specifically in the healthcare environment, by objects or materials that carry infection or transmission from individuals who have been infected [36,37]. The nasal passages have traditionally been considered the primary location of *S. aureus* colonization. However, *S. aureus* can inhabit various cutaneous areas and, in addition to the intestinal tract [38]. Sustained colonizing takes place in a small percentage of the population, varying from 10 % to 30 % based on the type of study. It is due to repetitive touch, nasal picking, and subsequent spreading [39].

S. aureus may be additionally transmitted through animals, particularly in the livestock sector, where the emergence of livestock-associated MRSA (LAMRSA) has become an increasing cause of concern [40]. The widespread *S. aureus* infestation always relies on the bacterial

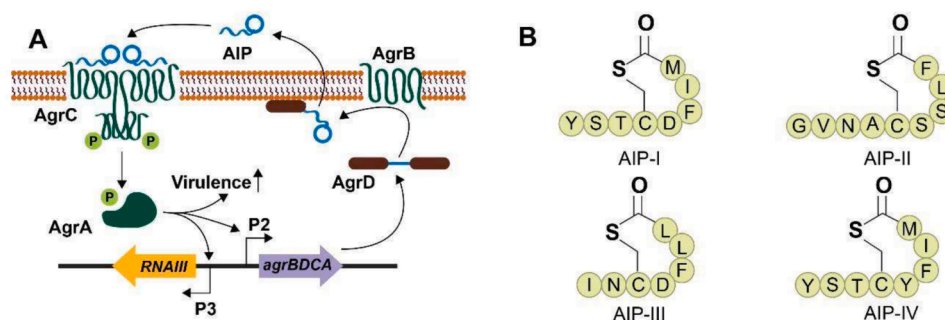


Fig. 1. Schematic representation of *S. aureus* *agr* system.

penetration of the epithelial safeguarding layer [41].

For instance, skin diseases may arise from small skin scratches and become infectious. Nevertheless, *S. aureus* can proactively induce epithelial rupture where the α -toxin has been mainly attributed to the initiation of metalloproteinase (Aur) domain-encompassing protein 10 (ADAM10) for the cleavage of E-cadherin molecules [42,43]. Disrupting attachment junctions jeopardizes the actin cytoskeleton [44], as shown in (Fig. 2).

There are two primary ways that the primary *S. aureus* toxin, α -toxin, functions. They rely on the ADAM10 receptor, which has disintegrin and Aur sites. Initially, α -toxin creates a heptameric pore that triggers pore development in an array of targeted cells. Furthermore, the compromise of the cytoskeleton breaks adhering junctions, resulting in epithelium and endothelium rupture [45]. The cross-contamination of implanted healthcare supplies indicates an additional source of spreading *S. aureus* infection in hospital environments.

The primary driver behind this infectious pathway is the ability of *S. aureus* to attach to the plastic component of the implants and the matrix molecules that coat them immediately after inserting and developing biofilm on the device [46]. The developed biofilm is also responsible for menstruation staphylococcal toxic shock syndrome (TSS), where certain *S. aureus* species generating TSS toxin-1 (TSST-1) build biofilms on high-absorbing menstrual products [47,48].

Food-borne illness is a sudden *S. aureus* infection that occurs when food items are adulterated by staphylococcal enterotoxins (SEs) during consumption [49]. SEs induce histamine production from intestine immunological cells of the myeloid lineage [50]. Like TSST-1, these superantigen toxins primarily induce unspecified stimulation of T cells, leading to a dominating immunological response that involves enormous cytokine production and polyclonal stimulation of T cells [51]. Additionally, *S. aureus* can opportunistically take advantage of the main damage caused by additional infections or predisposing circumstances. It happens, for instance, in pulmonary illnesses brought on by infectious viruses like the flu, when a subsequent *S. aureus* invasion is frequently a contributing factor to fatalities [52].

In addition, *S. aureus* was identified as a contributor to the emergence of atopic skin conditions by stimulating mast cells through particular toxins such as δ -toxin or associated cytolytic peptides known as phenol-soluble modulins (PSMs) [53–55]. Furthermore, *S. aureus* can exacerbate dermatological infections triggered by other pathogenic agents. An aggravation of Buruli ulcers following the initial infection by the pathogen *Mycobacterium ulcerans* had been eliminated by antibiotic therapy [56].

2. Antibiotic resistance in *S. aureus*: An increasing threat

S. aureus, a common invasive Gram + ve bacteria, presents a serious problem in the twenty-first century, creating financial, medical, and healthcare-related problems. Patients who are hospitalized in an intensive care facility and have medical gadgets in their bodies are more vulnerable to *S. aureus* infestations [57]. Current findings indicate that *S. aureus*, specifically MRSA, has developed resistance to traditional second and third-line antibiotic treatments. Understanding drug-resistant mechanisms in *S. aureus* is crucial to preventing the infection from developing novel resistance and spreading to other species [58].

In 2021, worldwide survey on antimicrobial resistance (AMR), the World Health Organization (WHO) voiced concerns over resistance established by bloodstream-associated disease-causing infectious agents, such as MRSA. It is essential to comprehend the process of resistance that pathogenic microbes have constructed since this may be investigated to battle resistance formation successfully [58,59]. Recurrent administration of various drugs induces a selection pressure that leads to the emergence of MDR and increases the risk of MRSA strains developing. The primary causes of this resistance are the β -lactamases enzymes they carry, the stimulation of efflux pumps, and mutations or modifications in the targetted area [60].

MRSA strains are spread by horizontal transmission of genes from transposons, resulting in the emergence of genes and virulence factors like *mec*. The *mec* gene initiates the resistance towards Methicillin in the genome islands SCCmec (staphylococcal chromosomal cassette *mec*).

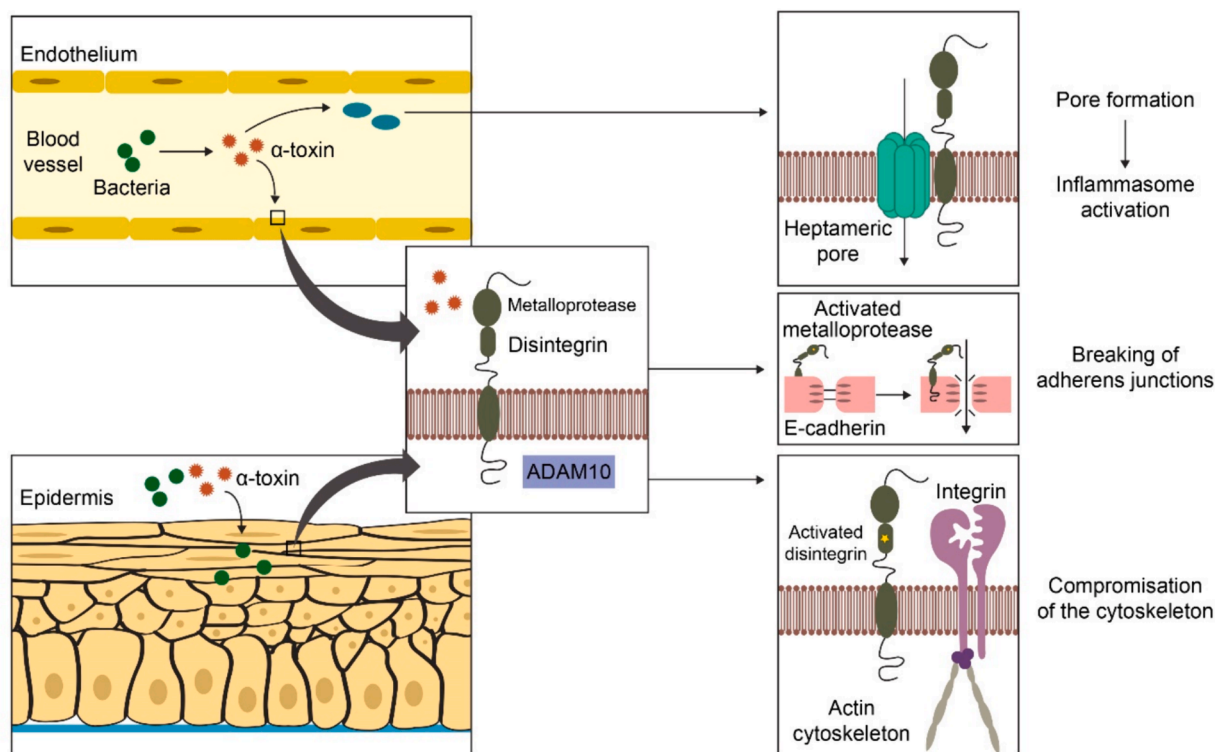


Fig. 2. Effect of α -toxin in *S. aureus* infestation.

After that, IcaB controls PIA's partial acetyl group removal to boost positive charge and enhance adherence. Lux can inhibit *rbf* expression and accelerate the biogenic synthesis of AI-2, which increases the levels of expression of *icaR*, lowers the transcript levels of *icaA*, and inhibits the creation of PIA-associated biofilms [81]. In strains with *luxS* mutations, there was an upsurge in *rbf* expression. Moreover, *rbf* gene expression could return to normal when supported with *luxS*-encompassing plasmids or efficacious doses of exogenous AI-2. The findings imply that *luxS* may inhibit *rbf* expression and lower *icaA* transcript levels via AI-2-assisted signaling, leading to reduced PIA-based biofilm development [82]. Therefore, the LuxS/AI-2 QS mechanism may be a competing target in intercellular interaction.

3.2. Agr system

The Agr system extensively investigates the *S.aureus* controlling mechanism concerning the QS approach, its gene regulated by the regulatory molecule and targeted gene controlling processes [72,83]. The Agr system in *S. aureus* comprises transcripts RNAII and RNAIII, initiated by the promoters *P2* and *P3* [84]. *S. aureus* *agr* has four significant protein-rich subunits ribosomally transcribed from one transcript and situated on a solitary chromosome locus [85]. AgrD is an approximately 45 peptide remnant upon cleavage and cyclizes by the transmembrane endopeptidase AgrB. The QS circuitry comprises an operon of four genes, *agrA*, *agrB*, *agrC* and *agrD*, as shown in (Fig. 4). They are coded next to a regulating RNA replicated in reverse order and control most of the Agr regulon [86].

The arrows in red indicate the various processes associated with Agr-QS and targeted gene regulation. AgrB alters and exports the AgrD AIP precursors, then proteolysis trimming by MroQ. When AIP reaches a certain threshold level, it results in post-translational modifications of the AIP sensor membrane AgrC, which subsequently phosphorylates AgrA. AgrA initiates the gene expression process from the *agrBDCA* promoter and the promoter regulating transcription of RNAIII, a regulator sRNA that regulates Agr transcription of the targetted gene. AgrA

promoter attachment exerts outstanding immediate regulation over phenol-soluble modulins (PSM) genes. Ambuic acid drug found to hinder AIP production (AgrB) [87].

AgrB is the enzyme that transforms the *agrD* gene expression after gene translation to produce a thiolactone ring. There are four distinctive Agr groupings, each with significant changes in the arrangement of the amino acids and, to some degree, the matured autoinducing peptide (AIP) length [88,89]. The AIP can be seven or nine amino acid residues, which AgrB transfers and further trimming at the *N*-terminal region with the assistance of non-Agr-coding membrane positioned protease MroQ for AIPs groups 1, 2, and 4. AIP types I and III are categorized as groupings II and III because of their extremely varied amino acid sequences. On the other hand, types I and IV are categorized as group I/IV, constituting three AIP groups, because they vary by only a single amino acid [90]. Each of the three AIP groups effectively impedes their respective functions. This mechanism is referred to as bacterial intervention [91].

In contrast, the proteolytic maturity of the output of the AgrB thiolactone introduction phase remains unclear for subcategory 3 [92,93]. The AIP initiates the AgrC-AgrA system via adherence to the histidine kinase membrane AgrC, resulting in AgrA phosphorylation, which is a DNA-attaching response controller that interacts and triggers the promoters initiating transcription of *agrBDCA* and RNAIII [94]. AgrA also adheres instantly to the promoter of PSM genes, which Agr remarkably manages specifically compared to other Agr-controlled genes that regulate through RNAIII [95].

The QS phenotypic is initiated during the beginning by the limited action of the *agrBDCA* locus, which is significantly increased by the QS feedback cycle with AIP buildup in a highly distributed bacterial culture [96]. The primary biological motive for this regulation in *S. aureus* is thought to be to prevent the generation of toxins and generated aggression enzymes till the bacterial infestation has progressed to the point where nourishment is limited, tissue deterioration is required, and when the pathogen is capable of resisting the host's inflammation defense processes initiated by these influencing variables [97,98].

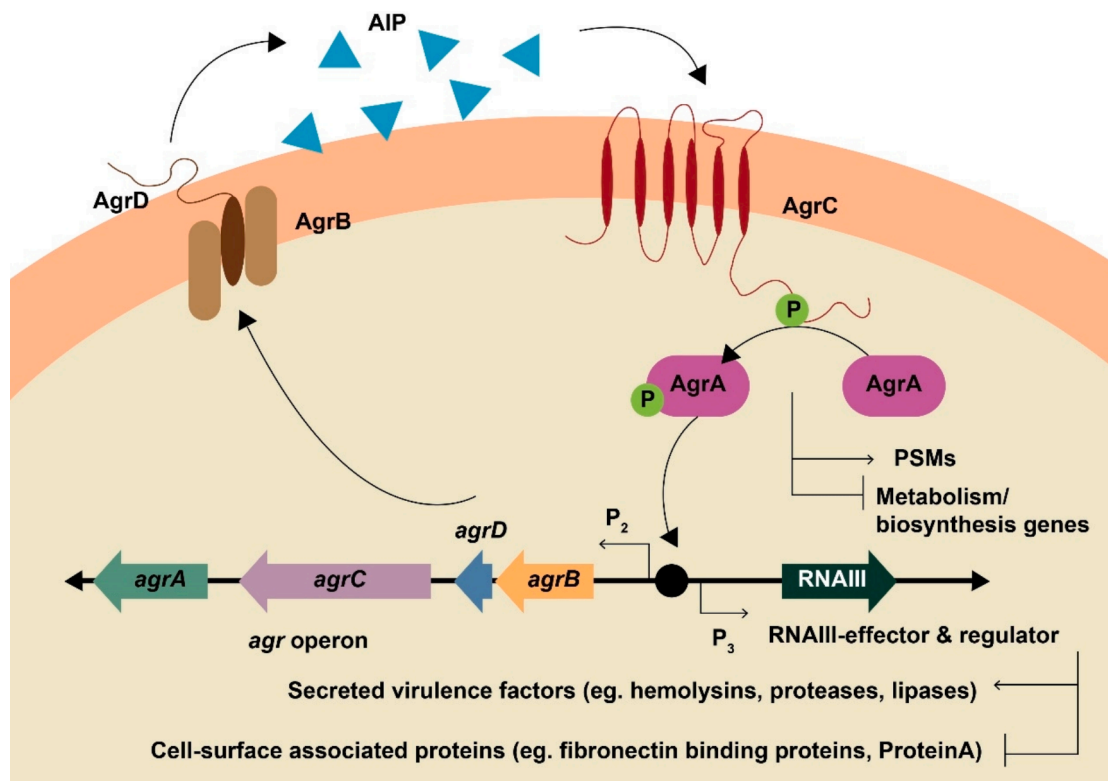


Fig. 4. The Agr circuit and diverse mechanisms underpinning the Agr-QS system.

3.3. SarA (Staphylococcal accessory regulator)

SarA is a crucial transcription controller that regulates the expression of several virulence-related genes and initiates the *agr* operon [99]. SarA is a DNA-adhering protein coded by the *sarA* locus [100]. The prototype SarA and its homology genes, SarR, SarT, SarS, MgrA, and Rot, comprise the SarA protein family [101], as shown in (Fig. 5). These proteins regulate the expression of various genes, mostly by controlling transcription but occasionally through post-transcriptional processes [102]. SarA is coded by the (*sar*) locus, consisting of three overlapping transcription units (*sarB*, *sarC*, and *sarA*) ending at the 3' end and starting from P1, P2 and P3 promoters [103].

The *sarA* locus is required for *S. aureus* biofilm development, PIA/Poly-β(1-6)-N-acetylglucosamine (PNAG) synthesis, and transcription of *ica* operon [104]. SarA controls biofilm development via the *agr*-associated mechanism, adheres to the *agr* promoter to initiate the transcriptional process of RNAIII, circuits and controls downstream targeted genes [105,106]. SarA regulates the transcription of genes by association with targeted promoters via the *agr*-independent mechanism. SarA initiates the P2 promoter and stimulates transcription by twisting the *agr* promoter DNA sequence. It enhances *agr* dimer associations, enhancing *agr* expression [107].

SarA harnesses its pleiotropy impacts on the transcription of genes by initiating the *agr* QS system adhering to P2 and P3 promoters of RNAII and RNAIII of *agr* loci [108]. SarA steadies regulating transcripts (*agrA* and *saeS*) and pathogenicity factor transcripts *hla*(α-hemolysin), *hlyB* (β-hemolysin) and *hlgCB* (δ-hemolysin) [109].

Nevertheless, one of the significant functions of SarA is its capability to suppress extracellular proteases (ECPs), such as aureolysin (Aur), which may destroy PSMs [110]. SarA upregulates exoprotein transcription and interacts with the Agr mechanism, suppressing ECP synthesis throughout biofilm development [111]. Based on transcriptome profiling, SarA regulates no fewer than 120 genes directly/indirectly. These include the increased regulation of the expression level of ECPs such as fibronectin-binding protein (FnBP), TSST-1, SEs and α-and β-hemolysins [112]. SarA functions as a transcriptional initiator, stimulating the *ica* operon's expression and the generation of PIA predecessor to initiate the formation of biofilms [113].

Rot is a key modulator of pathogenicity and biofilm-associated genes in *S. aureus*. It positively regulates ClfB, SdrC, and SarS, enhancing the surface level of proteins while decreasing extracellular enzyme

concentrations [114]. In *S. aureus*, RNAIII regulates Rot production by inhibiting its expression [115]. MgrA inhibits biofilm development by suppressing the synthesis of adhesins and functions as a negative controller [116]. SarA stimulates the expression of the *ica* operon and its inhibitor *icaR*, suggesting that it could hinder the overproduction of PIA [117]. Another contribution of SarA in biofilm development is the suppression of ECP synthesis. SarA genetic variations have a worldwide influence on the availability of various *S. aureus* transcriptomes synthesizing substantial quantities of proteases such as Aur, serine protease (SspA) and cysteine protease (SspB and SspC) and prevent the bacteria from developing biofilms [118].

Alternative sigma factor σ^B (SigB) represses several exoprotein genes essential for biofilm propagation [119]. σ^B , a *sigB* operon derivative, regulates *S. aureus*'s susceptibility to environmental stresses. It promotes bacterium resistance to drugs, biofilm growth, and pathogenicity-associated transcription of genes [120]. Osmotic stresses greatly enhance the *S. aureus* wild variant MA12's ability to produce biofilms; however, the *sigB* deletion mutation prevents this ability [121].

σ^B causes a rise in SarA and a reduction in Agr system RNA III levels, resulting in the development of phase-dependent variations in a few factors associated with virulence [122]. Additionally, it promotes the synthesis of many cellular surface proteins, including FnbA and ClfA, that are involved in the initial adherence of biofilms. It was discovered that terpenoid nootkatone therapies reduced the expression level of the *sarA* gene, which in turn reduced the synthesis of PIA and the development of biofilms [123].

4. Quorum sensing targets and gene regulation

4.1. Virulence factor expression

One important factor that encourages the emergence of drug resistance in *S. aureus* is the production of biofilms. According to reports, biofilm can give microbial populations substantial survivability benefits, boosting strain resistance by up to 1500 times [124]. Biofilm formation may be categorized into four phases: (i) attaching and adherence; (ii) agglomeration with extracellular matrix formation and bacterium multiplication; (iii) biofilm organizational form and maturity; and (iv) biofilm dispersal with cell dissociation [125,126] as shown in the (Fig. 6).

During the early adherence or agglomeration phase, *S. aureus* plankton cells adhere to living or nonliving substrates and are wrapped in a complicated encasing made of host biomolecules such as proteins [127,128]. When bacteria adhere to one another throughout tissue infectious diseases, they can accumulate in thick mucus (like cystic fibrosis) or on injured host tissues (such as cardiac valves, bones, or the dermis of serious injuries) [129–131].

The anchored or accumulated cells proliferate throughout the development and proliferative stage, producing an extracellular polymeric substance (EPS) essential for building 3-D biofilm frameworks [132]. EPS molecules, comprising polysaccharides, nucleic acids, proteins, and lipids, offer biofilm rigidity, directly influencing cells' survival settings [133]. During biofilm propagation, the biofilm becomes unstable in cells and agglomerates, which disperse into the surroundings, adhere to new sites, and cause subsequent biofilm illness. Emerging biofilm sites depend on the mobility potential of the separated bacteria, culminating in novel outbreaks in adjacent tissues, such as implanted biofilms producing bone disease, or more places, like separated biofilms, leading to inflammation in heart chambers [134–136].

Bacterial cells proliferate, and matrix complexities increase, causing physiologic diversity within the biofilm, defined primarily by nutrient levels and O₂ concentrations [137]. There are four metabolic phases for the cells developing in the biofilm: (i) oxygenated (enriched with nutrients); (ii) fermentation (found in the innermost layer that is deficient in nutrients and O₂); (iii) inactive (found in the O₂-deprived layer with a sluggish development and ineffective metabolic processes); and (iv)

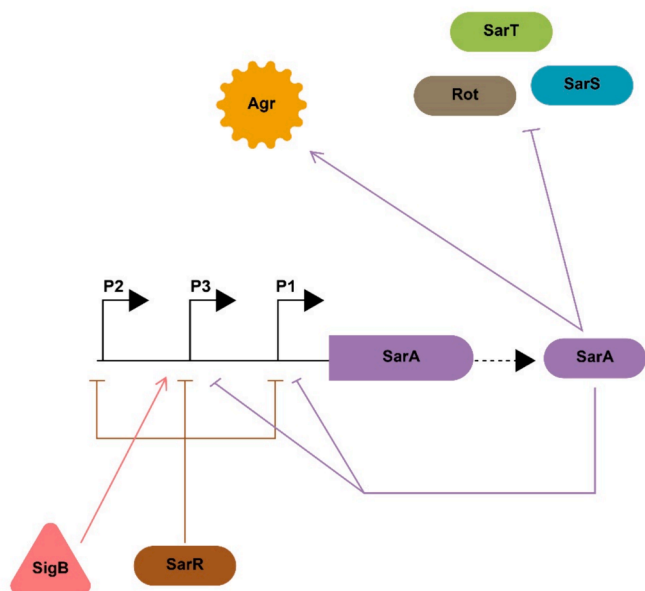


Fig. 5. SarA regulatory circuit and underpinning mechanisms.

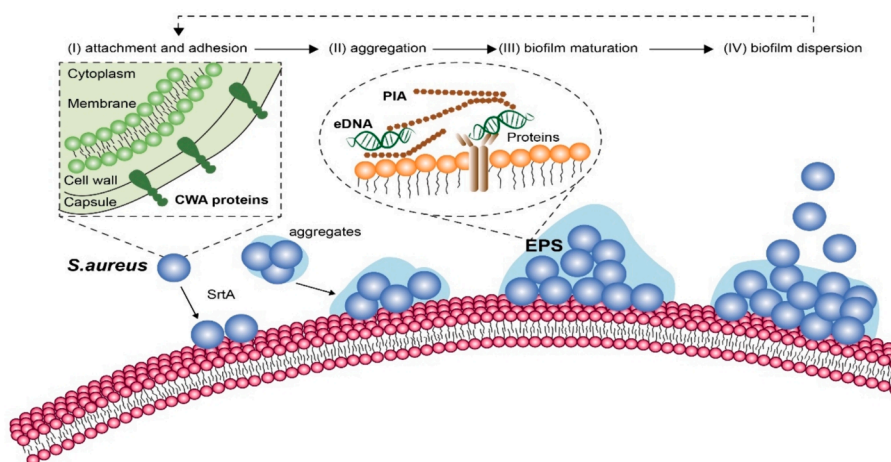


Fig. 6. Biofilm development mechanism in *S. aureus*.

dead [138–140]. A decrease in intracellular adenosine triphosphate (ATP) caused by dormant cells can reduce bacterial drug susceptibility [141]. A vertical gradation of viscoelasticity is created in the early phases of *S. aureus* biofilm growth. This gradient aids biofilm dispersion and eliminates weakly attached bacteria while maintaining an embedded layer inside the biofilm architecture [142].

4.1.1. Key components in *S. aureus* biofilm development

The EPS constituents for *S. aureus* biofilm comprise PIA, essential for multiple functions, including creating biofilm, antigenic escape, colony formation, and antibiotic resistance [143]. The synthesis of PIA is heavily influenced by environmental factors, including glucose and the absence of O₂ and is monitored by many regulatory mechanisms [144]. Most of the PIA-associated biofilm is seen in MRSA strains [145]. In this process, PIA provides the polymer with a +ve charge by removing the acetyl group, completing cell adherence and intercellular adhesiveness, and improving biofilm stability and drug resistivity [146]. Biofilm-based protein is associated with PIA to facilitate cell-to-cell accumulation during biofilm creation, and PIA functions together with the accumulated proteins to enhance the development of *S. aureus* biofilm [147,148].

During the initial stages of biofilm development, *S. aureus* cells mostly penetrate the host cells by adhering to their surfaces [149]. In this mechanism, *S. aureus* embodies many surface proteins covalently coupled with peptidoglycans, called cell wall-anchored (CWA) proteins. The most common CWA proteins comprising protein A, FnBPs, clumping facets (ClfA, ClfB), serine-aspartate repetitive group of proteins (SdrC, SdrD, and SdrE), fibrinogen binding protein (Fib) and *S. aureus* surface protein (SasG) promote the formation of biofilms by making it easier for *S. aureus* cells to adhere to EPS and host cells [150].

S. aureus produces four shortened PSMα peptides (about 20 amino acids), two extended PSMβ peptides (approximately 40 amino acids), and an RNIII-coded δ-toxin [151]. PSMs were recently reported as factors of *S. aureus* pathogenicity and possible peptides that promote inflammation reactions [152]. With the stated surfactant-like qualities, PSMs assist biofilm structure in forming pathways that provide nourishment accessible to *S. aureus* in the deep layer of the biofilm and encourage biofilm dissociation to liberate bacteria [153,154].

In *S. aureus*, PSMs are dissoluble and may assemble into amyloid fibers to sustain the biofilm framework [155]. As a result, PSMs could serve as an intriguing research area.

4.2. Impact on antibiotic resistance

4.2.1. Influence on efflux pump regulation

Efflux pumps (EPs) are protein-based carriers of pathogens attached

to membranes that expel drugs, reducing drug buildup and resulting in resistance to therapy, as shown in (Fig. 7). Their involvement in bacteria's physiology is to produce virulence determinants and extrude toxic compounds, giving a perspective that the medicines are immediate substrates of these transporters [156]. MDR EPs are microbial efflux systems that can transfer many chemotherapeutic medications or one category of antibiotics.

The picture represents a schematic illustration of the five architectural classes of antibiotic carriers. ABC: ATB-binding cassettes, MFS: major facilitator superfamily, RND: resistance-nodulation-division, SMR: small multi-drug resistance, MATE: multidrug and toxic compound extrusion. These five groups are further classified into two independent types: primary exporters, i.e., ABC transporters employing energy from ATP's hydrolysis process, and secondary exporters, i.e., MFS, RND, MATE, SMR groups enhanced by proton or Na⁺ ion gradient [157]. MATE group transporters use proton motor force (PMF) and Na⁺ membrane gradient for pumping out their substrate, while MFS, RND, and SMR exporters obtain their energy for drug extrudates from PMF; similarly, the ABC group uses ATP for drug extrudate mechanism [158]. In illnesses linked to biofilms, deactivating the EPs can be a helpful strategy for eliminating biofilms [159]. Various chemical substances or environmental stressors may cause the enhanced function of these resistive EPs, and this is one of the main theories explaining why drug-resistant *S. aureus* is more persistent [160].

4.2.2. Efflux pump inhibitors (EPIs)

On a global scale, resistance to medications in infectious agents is increasing due to the introduction of new resistant models. The approach used to combat drug resistivity is to investigate novel, prospective therapeutical medicines or improve the molecular pattern of existing drugs to evade resistance processes [161]. EPI is a promising strategy to counteract drug resistance by increasing medication effectiveness through aggregation within the cell. Numerous substances that hinder the EPs are recommended because their essential function interferes with efflux activity. These architecturally varied derived products of efflux interrupters work in tandem with most antibiotics commonly employed to boost their efficacy significantly [162]. In the intervening years, several molecules either developed, transformed or extracted from an indigenous source have been employed to increase the effectiveness of antibiotics towards infections [163].

Numerous EPIs have been identified from biogenic sources, but most of them, like reserpine, do not have therapeutic use because of their severe toxicity. Studies have identified a variety of architecturally multifaceted inhibitors of NorA, a multi-drug carrier of *S. aureus*, that conduct an action in an integrated way with the most extensively employed drugs and can boost their efficacy towards natural occurring

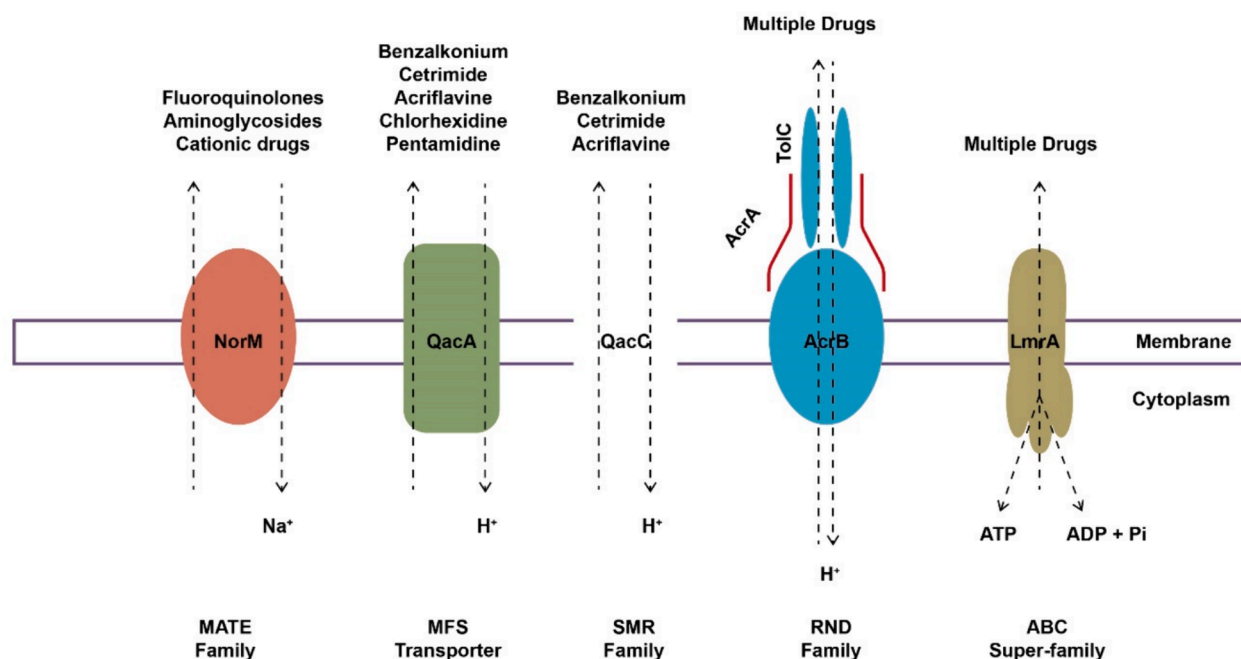


Fig. 7. Bacterial efflux pumps.

and NorA overexpression of *S. aureus* strains [164]. Biofilms, remarkably resistant to drugs, comprise extremely active EPs that transport and expel cell-signaling chemicals and antibiotics. Therefore, blocking the EPs by *EPI* may additionally hinder the production of biofilms and the QS system in cells [165]. Numerous investigations have shown that NorA is a crucial component of the *S. aureus* genome and that numerous *S. aureus* strains have been shown to express it [166].

The findings suggest that the QS mechanism and EPs are mutually dependent, and interference with one would either upward or down-regulate the other. Prior research has demonstrated a substantial relationship between QS, biofilm development, and multidrug EPs [167,168]. Certain microbes within biofilms develop antibiotic resistance due to processes such as the increased expression or availability of numerous gene-encoded EPs. Research revealed that plasmid interactions more often occur in biofilms than planktonic bacteria, raising the risk of biologically occurring and antibiotic-triggered resistance [169]. It has also been shown that inactivating bacterial EPs reduced QS signal release and biofilm development.

EPs activities are necessary for reacting to QS signals since the expulsion of these signals relies on the bacterial primary EPs system, and any undesirable substance generated inside is released through it [170]. In addition, in mixed-culture systems, efflux pumps function as a defensive system when QS inhibiting agents are used, preventing the passage of inappropriate signals from competitors. As a result, efflux pumps and QS impact each other's activities and their activity in response to the outside of the cell environment [171].

EPs have a role in both antibiotic distribution and cell communication in QS. They are demonstrated to be AIP secretors and hence seem to be a significant focus for antimicrobial therapy [172]. The presence of anti-resistive arsenals such as EPs and QS inhibitors and their influence on biofilm suppression have been fully considered while selecting innovative antimicrobial medicines targeting biofilm-based illnesses [173]. As a result, the combination of *EPI* with the QS system might be an important approach to eliminating biofilms [174]. These molecular classes provide the most important antibacterial armories for regaining drug effectiveness (*EPI*) and destructing biofilm formation, as illustrated in Table 1.

Table 1

List and architecture of compounds with inhibiting properties of biofilm and efflux pump activities.

Name	Reference
Chlorpromazine	[175]
1-(1-Naphthylmethyl) piperazine (NMP)	[176]
Carbonyl cyanide-chlorophenyl hydrazine (CCCP)	[177]
Phenylalaninearginine b-naphthylamide (PABN)	[178]
4',5'-O-dicaffeoylquinic acid	[179]
Thioridazine	[180]
Reserpine	[181]
Piperine	[182]

5. Environmental factors affecting QS in *S. aureus*

Various environmental and metabolic parameters, including pH, reactive oxygen species (ROS), and nutrient accessibility, may influence *agr* quorum sensing in *S. aureus*, as shown in Table 2.

QS in *S. aureus* is influenced by various environmental factors, which fall into three distinctive categories, as shown in (Fig. 8). (i) inhibition of QS via ROS activity produced by inherent immunity cells on *agr* system components, (ii) *agr* expression is either up- or down-regulated transcriptionally due to different non-*agr* protein regulators adhering to the *agr* P2-P3 promoter domain, and (iii) the environmental parameters that either up- or down-regulate the expression of *agr*, while the precise approach of action is still undetermined [202].

5.1. Influence of serum and ROS on *agr* components

Investigations of worldwide variations in *S. aureus* expression of genes reported that decreased transcription of *agr* RNAIII constituted noticeable impacts on development in fresh serum. Hence, serum includes at least a single QS suppressive factor [203]. Several streams of supporting information indicate that this quorum-quenching action, if not completely, is caused by ROS produced by innate immune cells [204]. *AgrA* and AIP are susceptible to ROS-induced oxidative stress. The effect of oxidative stress damages *AgrA*'s capability for attachment to the P2-P3 promoter region by forming disulfide bonds across Cys-199 and Cys-228, affecting its functionality [205]. ROS can disrupt QS in AIP

Table 2Environmental factors affecting *agr* function.

Host/ Environmental parameters	Influence	Process	References
Serum lipoproteins	Inhibition	Sequestering 4 types of AIP	[183,184]
SrrAB	Inhibition	SrrAB adheres to <i>agr</i> P2-P3 promoter area at minimum oxygen (O ₂) levels	[108,185]
AirRS	Inhibition	Lowered DNA adhering function of AirR at minimal O ₂ level hinders <i>agr</i> by undefined process	[186,187]
CodY	Inhibition	CodY adheres <i>agr</i> P2-P3 promoter region when nutrients are abundant	[101,188]
Sar A	Enhancement	SarA adheres to the <i>agr</i> P2-P3 promoter region and causes DNA bends, facilitating transcription.	[103,189]
SarR, SarX	Inhibition	Competition with AgrA for adherence to <i>agr</i> P2-P3 promoter region	[190,191]
MgrA, SarZ, and other candidates of the SarA group	Inhibition or enhancement	Regulate the expression of SarA, SarR, and SarX	[99,192]
Glucose	Inhibition	Inhibition due to the reduction in pH owing to catabolic activities	[193,194]
ROS (Reactive oxygen species)	Inhibition	Underwent oxidation of AIPs I and IV	[195]
ROS	Inhibition	The oxidizing lipid portion of lipoproteins leads to enhanced AIP sequestering.	[196,197]
ROS	Inhibition	Cys oxidation in AgrA prevents DNA adherence.	[198,199]
ROS	Inhibition or Enhancement	Alteration of sustained Cys in the SarA group of controllers affects the DNA attachment capability.	[200,201]

via two distinctive processes: one for AIP types I and IV and another for all four forms. Exposing AIPs I or IV to macrophage-derived ROS, whether within the living or artificial environment, might oxidize the adjacent chains of the C-terminus methionine, preventing the pheromones from activating AgrC [206].

Nevertheless, the identification of *agr* expression by either real-time polymerase chain reaction (PCR) assessment of RNAPIII proportions in an *agr* category I strain or by green fluorescent protein (GFP) expression through the regulation of the PSM promoter in an *agr* type III genus would signify ROS hindrance of AgrA or AIP for bacterium situated in polymorphonuclear neutrophil (PMN) phagocytes is of lesser significance [207].

The expression level of RNA III in comparison to glyceraldehyde-3-phosphate dehydrogenase (GADPH), referencing gene enhanced in the multiplicity of infection (MOI), demonstrating that at greater levels of MOIs, the bacterium could be able to outcompete NADPH oxidase (Nox2) capability to produce enough ROS [208,209]. More significantly, there was an 8–37 fold rise in RNAPIII expression compared to the opsonization of *S. aureus* controlling bacterium when NADPH oxidase was suppressed by pretreatment of PMNs with diphenylene iodonium. It implies that the *agr*-inhibiting action of serum may be attributed to ROS-suppressing AgrA or AIP processes. Another approach to serum inhibition is the sequester of AIP by lipoproteins. At the initial stage, a human serum pool was shown to disrupt QS by an *agr* type I strain while in vitro

development. Further reporter tests pinpointed the cause of suppression of low-density lipoprotein (LDL), very low-density lipoprotein (VLDL), and the apolipoprotein B (ApoB) portion of lipoproteins [210].

Surface plasmon resonance (SPR) demonstrated that pure ApoB binds directly to embedded AIP-I, suggesting that serum inhibits QS through an AIP-I sequester. ApoB adherence to anchored AIP-I was inhibited by easily soluble AIP-II and AIP-IV, as determined by SPR. It suggests that ApoB's sequester of AIP may apply to all four kinds [211]. The significance of ROS produced by Nox2 in the AIP sequester process was identified during subsequent investigation that evaluated the application of this process to AIP-III-dependent QS. It was identified that LDL hindered QS in an *agr* type III strain to a lower level than the pure form of ApoB, suggesting that architectural variations among unbound ApoB and ApoB in LDL might affect ApoB's binding capacity to AIP-III [212].

Since it was recognized that the oxidation of the lipid portion of LDL would change ApoB's configuration in the LDL, it was determined that oxidized LDL (oxLDL) had the same inhibitory effect on AIP-III-dependent QS as pure ApoB. Following that, it became apparent that greater vulnerability to becoming infected by an *agr* type III bacterium was associated with a lower level of oxLDL in the serum of Nox2 knocked-out mice [213]. SPR tests demonstrating the instantaneous attachment of oxLDL to anchored AIP-II or embedded AIP-IV proved the generic application of the AIP sequesters. Recent research suggests apoB100 in VLDL and apoB48 in chylomicrons may hinder QS by AIP sequester [211]. In *S. aureus* pneumonia paradigm, higher ApoB concentrations in the respiratory tract led to greater mortality and inflammatory conditions, emphasizing the significance of AIP sequester by serum lipoproteins as an intrinsic defense system [214].

5.2. Significance of oxygen and ROS via non-*agr* regulators

The accessibility of O₂ and ROS produced by innately immune cells can regulate transcription from the *agr* P2-P3 promoter through alterations in the capability of non-*agr*-controlling proteins to adhere to this promoter locale [215]. The AirSR and SrrAB subsystems integrate each other to suppress *agr* expression upon low O₂ accessibility, achieving redox regulation [216]. Long-term susceptibility to O₂ or oxidant stress damages the 2Fe-2S clustering in AirS, which is necessary for its kinase activities. It changes the level of phosphorylation of AirR and, consequently, its DNA-adhering capacity [217]. Although SrrB's action is susceptible to redox control, it is unclear how precisely kinase function is regulated. SarA proteins can help regulate *agr* expression through redox control [218]. SarA proteins have a wing-like helical shape and DNA binding action.

SarA group proteins (SarA, SarR, and SarX) adhere to the *agr* P2-P3 promoter locale, either hindering (SarR and SarX) or enhancing (SarA) expression by modifying its architecture. Other members, such as SarZ and MgrA, either facilitate or hinder *agr* expression by controlling the level of expression of the SarA family, adhering to the promoter locale. All SarA families have a sustained cysteine that may be oxidized or nitrosylated, altering its DNA adhering capability [219]. The SarA family of proteins can regulate *agr* transcription by oxidation or nitrosylation stress, an inherent immune system response, or nutrient metabolic settings. Moreover, phosphorylation of cysteine by kinase Stk1 demonstrates a process by which cell wall-targeted antibiotics may regulate *agr* expression by suppressing kinase action of Stk1 by antibiotics [220].

5.3. Influence of nutrients via non-*agr* regulators

Nutrient and development stage regulation of *agr* expression predominantly exists throughout the pleiotropy CodY repressor [221]. CodY assists Gram + ve bacterium in becoming accustomed to impoverishing conditions [222]. CodY adheres to branched-chain amino acids (BCAAs) and Guanosine triphosphate (GTP) under favorable nutritional

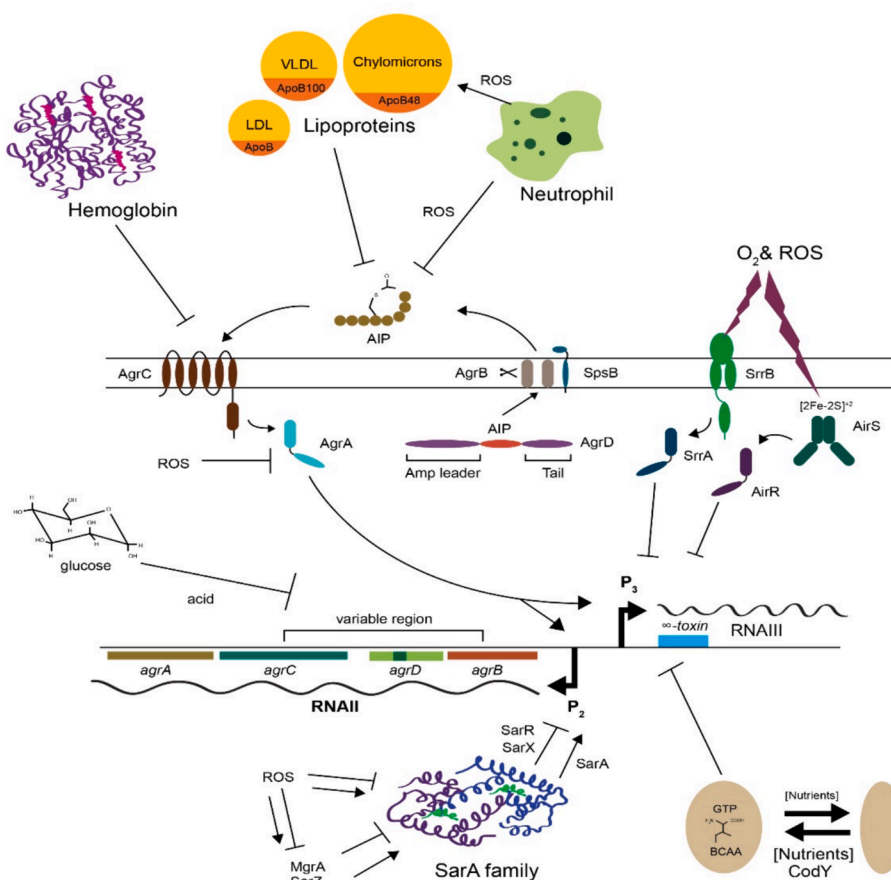


Fig. 8. Diagrammatic representation of the *agr* system in *S. aureus*, showing host and environment-related factors demonstrated to modify *agr* activities.

circumstances, inhibiting transcription by interfering with RNA polymerase and impeding elongation [223]. If nutrients are exhausted, BCAAs and GTP are incapable of adherence to CodY, causing CodY to become unable to attach DNA and suppression of transcription is alleviated [224]. Deleting CodY from the *agr* locus leads to increased transcription, whereas overexpressing it leads to reduced expression, as anticipated from a CodY-regulated locus. CodY adhering sequence was found in the *agr* P2-P3 promoter locale, suggesting its role in regulating *agr* expression throughout the developmental stage [225,226].

5.4. Influence of other factors via unidentified processes

The presence of numerous environmental variables, through unclear or unidentified processes, lowers the expression of the *agr* loci. Among them are glucose, iron and hemoglobin (Hb). The inclusion of heat-inactivating serum boosted *agr* expression. Adding Ferric chloride (FeCl_3) to the medium prevented the transcriptional rise. It suggests that the typically lower Fe levels in serum may favor *agr* transcription, and further addition of Fe inhibits *agr* transcription [227]. It was determined that adding Hb, globin chains/peptides sourced from Hb chains might decrease *agr* expression [228]. What mechanism underlies the inhibition of *agr* transcription by peptides generated from globin is uncertain.

It is widely recognized that the presence of glucose inhibits the gene expression of the *agr* gene; nevertheless, it is noteworthy to note that this suppression did not disappear upon the elimination of catabolite control protein A (CcpA), which is under the absence of a catabolite-responsive component at the *agr* loci [229]. Glucose-based suppression seems to be caused by a reduction in pH that is linked to the catabolic activity of glucose, as inhibition can be lessened by buffering the growing medium or nurturing the bacteria in a fermenter where the level of glucose and pH can be maintained consistently [230]. Moreover, it was identified

that lowering the pH of the growth medium during the shortage of glucose can imitate the inhibition seen when glucose is present.

Low pH connected with acidic phagocytes has been considered to suppress *S. aureus* exoprotein expression (*SaeRS*) in specific *S. aureus* strains [231]. Comparable to SarA, SaeRS decreases the post-translational disintegration of PSMs by inhibiting the level of expression of *aur* [110]. This control happens via a SarA unassociated process, which relies on functioning *SaeS*. Moreover, temperature significantly affects *S. aureus* gene transcription.

For example, genes coding secretory proteases and toxins (*aur*, *exsA*, *sspA*, *sspB* and *agrD*) showed higher transcription abundance at 34 °C than at 37 °C and 40 °C [232,233]. The Biological Research in Canisters-23 (BRIC-23) findings may be more pertinent to the circumstances in which *S. aureus* experiences the exterior to the host (such as development/permanence on passive vectors, which may affect *S. aureus* propagation) [234].

6. Inhibition of QS in *S. aureus*

As pathogenic bacteria become more resistant to antibiotics, research has switched towards developing new tactics to eradicate them. Targeting QS is a potential technique, as it has been proven that QS has a crucial role in disease pathogenicity. Inhibiting QS, also known as quorum quenching (QQ), prevents bacteria from producing signaling chemicals that facilitate interaction [235]. QQ aims to impede cell interaction in a community rather than kill the pathogen. The method involves inhibiting the pathogen's capacity to detect the cell population, which hinders the initiation of gene expression [236].

RIP(RNAIII-hindering peptides) impedes RNAIII production in the RAP-TRAP (target of RNAIII-initiating protein) QS synthesis. RIP is a heptapeptide (YSPXTNF-NH₂) that reduces the infectious potential of

S. aureus, where X denotes a Cys, Trp, or an altered amino acid [237]. RIP intervenes with RAP from phosphorylating TRAP, which inhibits transcription from the P2 and P3 promoters and toxin production. YSPWTNF, the engineered analog of RIP, has been shown to reduce RNAIII production in vitro and inhibit *S. aureus* infections produced from multiple variants in vivo, including infectious mastitis, arthritis, keratitis, and cellulitis. Preventing the RAP-TRAP association in vivo model systems, either on its own or in conjunction with antibiotics, may effectively reduce drug-resistant *S. aureus* infections, similar to the *agr* system. RIP variants were created and studied as potential therapies for combating *S. aureus* infestations [238].

Effective disruption of *agr* QS in infectious agents reduces their virulence and ability to cause illness. Various approaches to reduce virulent gene expression in the *agr* operon are illustrated in (Fig. 9). Stage 1 inhibits the catalytic activity of AgrB. In stage 2, natural QS antagonists and engineered AIP analogs can inhibit AgrC competitively. In stage 3, Inhibiting histidine kinase activities are inhibited. In stage 4, RNA III expression is suppressed, and in the final stage, AgrA – P2/P3 association is hindered.

AgrA may be a therapeutic target for suppressing QS, as it functions critically by preventing phosphorylated AgrA from adhering to the P2 and P3 promoters and suppresses the *agr* system, inhibiting the formation of pathogenicity factors such as α -hemolysin and PSMs. Hence reduces *S. aureus* pathogenicity [239]. Infectious agents affected in this manner have trouble occupying host tissues and are finally eliminated by the host's immune defense mechanism. AgrB and AgrD are responsible for the generation of AIP. Antagonists targeting AgrB or blocking AgrD release can effectively suppress *agr*.

Certain peptide counterparts can prevent the AgrD activation site's breakdown and cyclization process, damage the AgrB enzymatic framework, or render it unresponsive. Many investigations have been conducted to identify QQ agents that target the cyclic peptide-mediated QS. Bioactive molecules in prokaryote and eukaryote origins have been recognized as QQ agents. These chemicals have different mechanisms of action, inhibiting specific locations in the QS system [240]. Table 3 shows QS inhibitors, their origins, and potential targets in *S. aureus*.

7. Natural products that neutralize *S. aureus* infections

Naturally derived compounds with anti-toxin characteristics or targeting the *agr*-QS system may be promising alternative therapies for *S. aureus* infestations [256,257]. Since synergistic techniques have delivered successful medications, creating novel antibiofilm agents based on natural materials such as plant-based extracts and phytochemicals is anticipated to result in potential innovative biofilm-fighting tactics, as shown in (Fig. 10). Biologically active organic components such as staphylococcal toxin neutralizers (e.g., morin hydrate and oroxylin glycoside), as well as quorum quenchers targeted at the *agr*-QS system (such as ambuic acid, chrysin, isorhamnetin, naringenin, puerarin, ω -hydroxyemodin, 224CF2 and 430D-F5), are illustrated in Table 4 [258].

A monoterpene glycoside called paeoniflorin, isolated from Paeoniaceae plants like peonies, could lower the degree of virulence and luxS/AI-2 system-regulated biofilm growth [272]. It was discovered that biosurfactants derived from *Lactobacillus plantarum* and *Pediococcus acidilactici* interfered with AI-2 secretion [273]. Activating the Agr mechanism is an effective antibiofilm technique since it prevents biofilm development and dispersal. Brazilin, a bioflavonoid found in *Caesalpinia sappan*, has been shown to suppress biofilm development by affecting Agr-associated activity [274]. It has been demonstrated that *Ginkgo biloba* exocarp extraction suppresses the transcription of two MRSA biofilm-dependent factors, *sarA* and *sigB*, in *S. aureus* [275].

Scrophularia ningpoensis honey has been shown to reduce gene expression in *S. aureus*, including *icaA*, *icaD*, *sarA*, *agrA*, and *sigB* [276]. A naturally occurring isoflavonoid called biochanin A can halt biofilm development by blocking PIA discharge and causing the separated EPS matrix to break up the constructed biofilms [277]. Natural compounds such as luteolin, quercetin, and kaempferol from *Euphorbia humifusa* were found to reduce biofilm development by inhibiting these surface-adhesive proteins expressing genes (e.g., *fnbA*, *fnbB*) [278]. Numerous natural compounds have been studied for their ability to suppress membrane sortase A (SrtA). It has been found that kaempferol inhibited SrtA functionality and down-regulated the transcription of genes linked to CWA proteins, preventing the development of *S. aureus* biofilm [279]. Similarly, it has been found that chalcone, quercetin, and myricetin had significant inhibitory effects on SrtA [280,281].

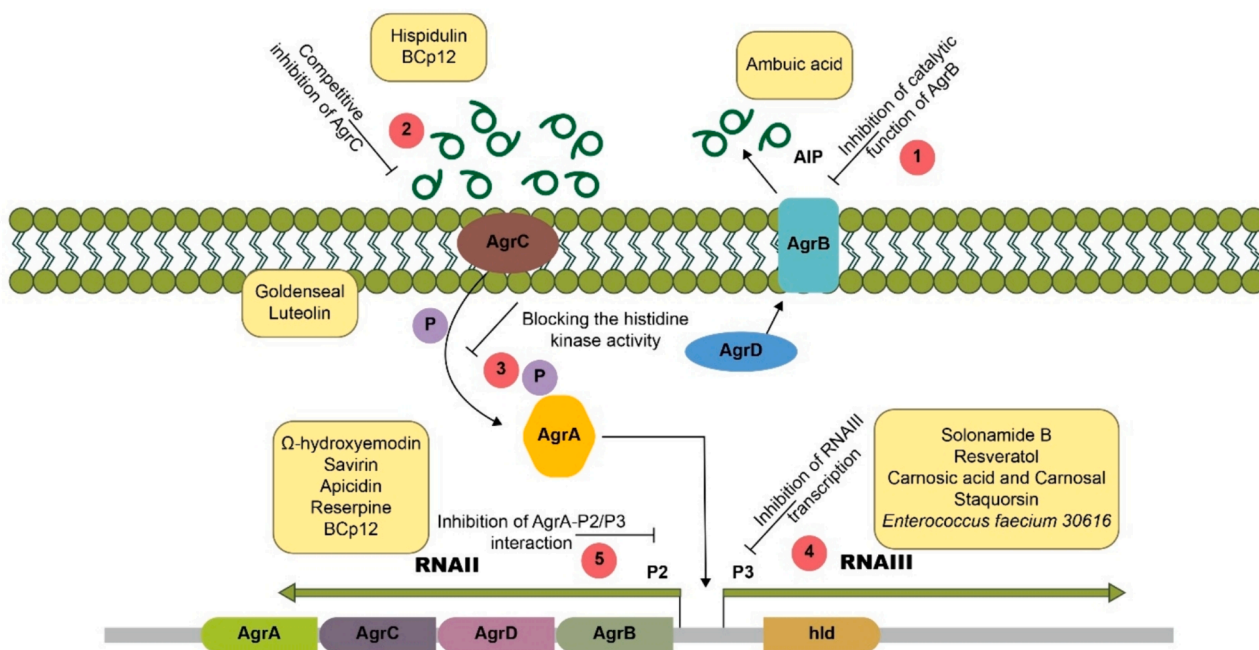


Fig. 9. QS inhibition mechanism, inhibitory agents and their target in *agr* operon.

Table 3
Quorum sensing inhibitors and their potential targets in *S. aureus*.

Quorum sensing inhibitor	Derived from	Mode of action	References
Avellanin C	<i>Hamigera ingelheimensis</i>	Impedes <i>agr</i> signalling circuit	[241]
Ambuic acid	Small molecule fungal metabolite	Inhibiting <i>agr</i> signal biological synthesis	[242]
Resveratrol	Fruits and nuts	Downregulating <i>hla</i> and RNAPIII	[243]
Carnosic acid and carnosal	Rosemary extract	Hindering of RNAPIII <i>agr</i> virulence expression	[244]
Bumetanide	Diuretics drug	Inhibiting AgrA	[245]
Staquorsin	4-Methoxy-N'-(phthalazine-1-yl) benzenesulfonohydradize hydrochloride	Downregulating α and δ hemolytic action, lipolytic function and downregulating RNA III transcript	[246]
Hispidulin	Plants	Hindrance of α -hemolysin by targeting AgrA _c	[247]
<i>Enterococcus faecium</i> 30,616	<i>Lactobacillus acidophilus</i> strain DSM 13241	Downregulating <i>hla</i>	[248]
Luteolin	Fruits and veggies	AgrA expression	[249]
Hamamelitannin (2', 5-di-O-galloyl-d-hamamelose)	<i>Hamamelis virginiana</i>	Impedes QS via TRAP receptor	[250]
Savirin	Triazoloquinazoline derivative	The specific target of AgrA	[251]
Extract 430D-F5	<i>Schinus terebinthifolia</i>	Suppresses the <i>agr</i> mechanism	[252]
Ω -hydroxyemodin	<i>Penicillium restrictum</i>	Hinders the association of AgrA with <i>agr</i> P2 promoter	[253]
Solonamide B	<i>Photobacterium halotolerans</i>	Decreases the transcription of RNAPIII	[254]
Goldenseal	<i>Hydrastis canadensis</i>	Targeting AgrC/A system	[255]

Table 4
Biogenic products that neutralize *S. aureus* infections.

Quorum sensing inhibitor	Derived from	Mode of action	References
Morin hydrate	<i>Maclura pomifera</i> , <i>Maclura tinctoria</i> , and <i>Psidium guajava</i> leaves	Hinders the self-assembly of Hla's heptameric transmembrane pore	[259,260]
Oroxin A Oroxin B Oroxilin A 7-O-glucuronide	Dried matured seeds, strawberries, grapes, apples, <i>Bignoniaceae</i> , onions and so on	Adhere to the central region of Hla and prevent the toxin from architectural transitioning from monomer to oligomer.	[261,262]
Isorhamnetin	Therapeutic herbaceous plants, as well as edible fruits such as apples, pomes, cherries and pears	Suppresses the expression of Hla and RNAPIII	[263]
Chrysin	<i>Passiflora caerulea</i> , <i>Passiflora incarnata</i> , <i>Oroxylum</i>	Impede RNAPIII and Hla expression	[264]
Puerarin	<i>Pueraria lobata</i> Ohwi roots	Hinders Hla and RNAPIII expression	[265,266]
Naringenin	Various fruits and herbaceous plants	Suppress the transcription levels of <i>agrA</i> and Hla	[267–269]
Hydroxyemodin (ω -OHM)	<i>Penicillium restrictum</i> fungi	Hinders <i>agr</i> functioning	[253,270]
Ambuic acid	Fungus-derived metabolite	Impede the synthesis of RNAPIII and AIP	[242]
224C-F2	<i>Castanea sativa</i> leaves	Restricts <i>agr</i> functioning and Hemolysin- δ (Hld, δ -toxin) synthesis	[271]
430D-F5	<i>Schinus terebinthifolia</i> berries	Restricts <i>agr</i> functioning and Hemolysin- δ (Hld, δ -toxin) synthesis	[244]

Emodin, an anthraquinone, a byproduct derived from *Rheum palmatum* and *Polygonum cuspidatum*, inhibited the formation of *S. aureus* biofilms by reducing the transcription of *cidA* and interfering with the discharge of eDNA [282]. Biogenic alkaloid Sinomenine, extracted from *Sinomenium acutum*, dramatically lowers the synthesis of PSMs, PIA, and adherence between cells, which are the main causes of *S. aureus* biofilm dispersion [283]. Sharifi et al. [284] studied the antibiofilm process of *Cuminum cyminum* essential oil (CcEO) on MDR *S. aureus*. CcEO at sub-MIC levels (0.6–1.3 μ L/mL) substantially decreased *hld* and *ica* expression by 3 and 2 fold, indicating its QS antagonistic ability.

The existing processes of natural therapeutics' antibacterial and antibiofilm activity are summarized in (Fig. 11). It should be noted that while nature-derived products are considered harmless, they are not always as safe as commonly believed. As a result, it is very necessary to examine its toxicity before a natural substance is recognized as a potential alternative in clinical therapeutic applications [285,286].

8. Novel therapeutic strategies to target QS in *S. aureus*

Anti-virulence strategies for *S. aureus* either focus on a virulence factor that is widely distributed and has an exceptionally significant effect on pathogenesis or attempt to eliminate many virulence indicators simultaneously [287]. Three key anti-virulence techniques for *S. aureus* are now being explored, each utilizing one or all of these pathways. The formation of MABs towards α -toxin is a common virulence factor in many strains, such as MEDI4893 (suvratoxumab), innovated by MedImmune [288,289]. Secondly, some methods target α -toxin and all *S. aureus* leukocidins.

The Austrian company Arsanis used this strategy to create mAbs that exhibit cross-reaction towards leukocidins and α -toxin [290,291].

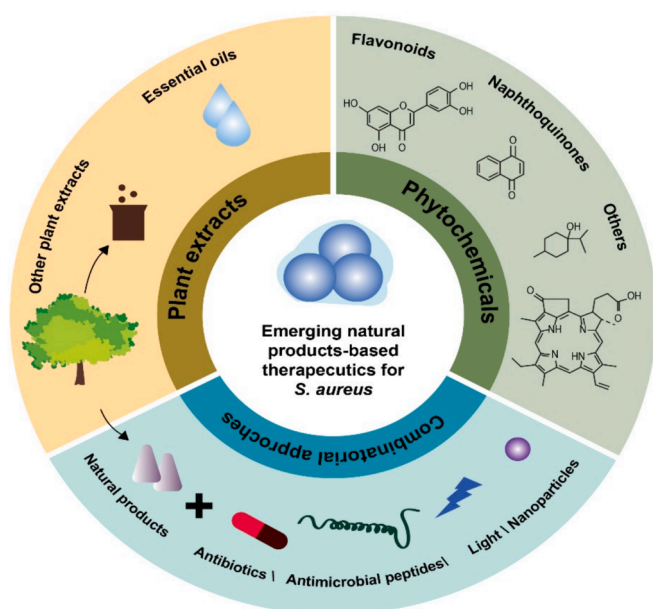


Fig. 10. Innovative natural-derived therapeutics to combat *S. aureus* infections.

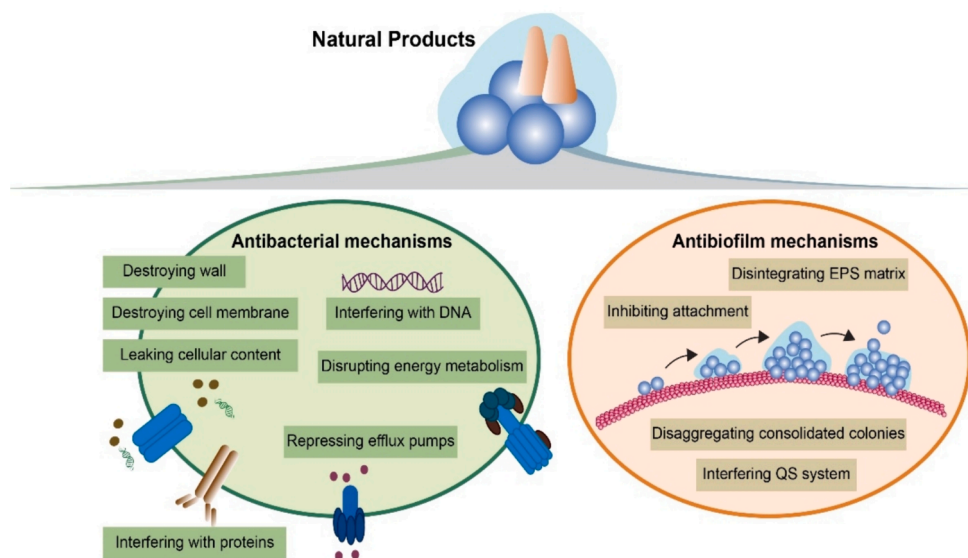


Fig. 11. Antibacterial and antibiofilm processes of nature-derived products.

Thirdly, various scientists have investigated QS inhibitors that specifically target Agr. They fall into two primary groups: those that impede AIP attachment to AgrC from outside the cell, like fengycins and AIP analogs, and those that must enter the cytoplasm to block the response regulator AgrA. These are typically more hydrophobic, which include savirin and apicidin [292]. Strong antimicrobial, anti-QS and antibiofilm properties are demonstrated by metal-based NPs derived from platinum (Pt), silver (Ag), zinc (Zn), and selenium (Se) toward drug-sensitive and resistant varieties of *S. aureus*. Their activity disrupted the bacterial membrane, allowing the release of intracellular substances [293,294].

A nanocarrier network is formed during the stretch and building of the hydrophilic and hydro-resistant nanoemulsions layers, followed by a thermally stable framework that may overcome infectious bacterium resistance [295]. The antibacterial functioning and effect on QS genes of *S. aureus* strains were evaluated employing the ascorbic acid nanoemulsions, and results exhibited notable defense towards *S. aureus* and hindrance of QS activity [296]. Chitosan-encased nanoemulsions are efficient in combating drug-resistant *S. aureus* [297].

Antimicrobial photodynamic therapy (APDT) is a potential way to combat antibiotic resistance and reduce reliance on commonly available drugs. APDT eliminates microorganisms by combining harmless photosensitizers (PS) with light radiation. To eliminate *S. aureus* infectious cells via in situ, the PS and molecular O₂ are exposed to certain light wavelengths that initiate ROS [298]. The processes include enhanced permeation, loss of potassium ions, cytoplasm membrane protein modifications, and cell wall breakdown. Photodynamic treatment demonstrated in vitro studies found effective against drug-resistant *S. aureus*, potentially replacing traditional antibiotics [299].

Li et al. [300] found that the antimicrobial peptide BCp12 derived from buffalo casein hydrolysate inhibited *S. aureus* biofilm development and investigated its potential mode of action using proteomics and molecular docking techniques. BCp12 showed considerable down-regulation of proteins of the *agr* QS system (AgrA, AgrB, AgrC, and PSM β). Hu et al. [235] created a bioactive hydrogel combining carboxymethyl chitosan(CMC)/sodium alginate with berberine, a QS inhibitory and antimicrobial agent. They used *Spirulina platensis*, a naturally occurring microalgae, to create a multifunctional hydrogel (BBR@SP). Upon laser illumination, the BBR@SP gel releases BBR and produces ROS, resulting in combinatorial QS inhibition toward chemophotodynamic treatment for MRSA. BBR@SP gel prevents biofilm growth and suppresses pathogenic factor expression. Their findings

proposed a new antibacterial strategy that lowers biofilm hypoxia, inhibits QS, and eliminates MDR bacteria.

9. Conclusion and future outlook

S. aureus-associated infections are rapidly spreading diseases linked to invasive and communal infestations. The pathogen's capacity to release viral components completely relies on the QS mechanism. In *S. aureus*, the most common QS systems are the *agr* operon, LuxS, and sarA. The Agr approach systematically examines the *S. aureus* regulating mechanism regarding the QS strategy, its gene controlled by the regulating molecule, and target gene control mechanisms. The LuxS/AI-2 QS mechanism may be a competitive communication strategy between cells. SarA, a key transcriptional regulator, controls the expression of various pathogenic associated genes and starts the *agr* operon. QS signaling mechanisms play a crucial role in bacterial pathogenicity, making them intriguing targets for drug discovery. Effective disruption of QS reduces their ability to spread illness. Hence, the establishment of prolonged virulence can be eradicated.

The bacterium's resistance to conventional drugs and the human immune system necessitates novel discoveries and strategies for inhibiting infection-causing agents. Among the tactics for targeting *agr* QS include inhibiting AgrA – P2/P3 interaction, limiting histidine kinase activities, and reducing AgrB's catalytic functionality. Various biologically active compounds derived from prokaryotes, eukaryotes and plant extract sources have been engineered for targeting *agr* QS systems. Despite identifying natural compounds with partially explained processes of action, most research was an observatory, necessitating a thorough investigation of pertinent processes in combating *S. aureus* infestations.

EPIs effectively combat pathogen resistance and improve therapeutic efficacy, making them a novel option for pathogen control. EPIs can enhance antibacterial treatment for pathogenic *S. aureus*, enhancing clinical findings. There is no doubt that the combination of EPIs and the QS cell signaling pathway has demonstrated a novel approach to reducing virulence in *S. aureus* by preventing the development of biofilms. EPs and QS in bacteria have a widely recognized relationship. Despite the preceding knowledge, an association between EPI and QS inhibitors requires further exploration to yield promising findings. Numerous investigations show that various environmental variables, including serum factors, oxygen, ROS, low pH, nutrition, glucose, and hemoglobin, can suppress the activity of the *agr* in *S. aureus*. The growth

within optimal experimental settings served as the baseline condition for these investigations. Better techniques for assessing *agr* QS kinetics in vivo would be beneficial for further studies on the function of *agr* QS in persistent biofilm infectious paradigms and colonized investigations. The novel techniques employed for targeting QS include the application of antimicrobial peptides, monoclonal antibodies, nanoemulsion, APDT, photodynamic therapy, and microalgae. Hence, an in-depth investigation is needed to validate their effectiveness against *S. aureus* infestations.

CCRediT authorship contribution statement

Kayeen Vadakkan: Writing – original draft, Visualization, Investigation, Conceptualization. **Kuppusamy Sathishkumar:** Writing – review & editing, Visualization, Investigation. **Sithara Kuttayahann Urumbil:** Supervision, Investigation, Conceptualization. **Sreejesh Ponnenkunnathu Govindankutty:** Validation, Supervision, Data curation. **Ajit Kumar Ngangbam:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Bijayalakshmi Devi Non-gmaithem:** Writing – review & editing, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] G.Y.C. Cheung, J.S. Bae, M. Otto, Pathogenicity and virulence of *Staphylococcus aureus*, *Virulence* 12 (2021) 547–569, <https://doi.org/10.1080/21505594.2021.1878688>.
- [2] N.A. Turner, B.K. Sharma-Kuinkel, S.A. Maskarinec, E.M. Eichenberger, P. Shah, M. Carugati, T.L. Holland, V.G. Fowler, Methicillin-resistant *Staphylococcus aureus*: an overview of basic and clinical research, *Nat. Rev. Microbiol.* 17 (2019) 203–218, <https://doi.org/10.1038/s41579-018-0147-4>.
- [3] P. Nikolic, P. Mudgil, The cell wall cell membrane and virulence factors of *Staphylococcus aureus* and their role in antibiotic resistance, *Microorganisms* 11 (2023) 259, <https://doi.org/10.3390/microorganisms11020259>.
- [4] M. Otto, Critical assessment of the prospects of quorum-quenching therapy for *Staphylococcus aureus* infection, *Int. J. Mol. Sci.* 24 (2023) 4025, <https://doi.org/10.3390/ijms24044025>.
- [5] A.P. Kourtis, K. Hatfield, J. Baggs, Y. Mu, I. See, E. Epton, J. Nadle, M.A. Kainer, G. Dumyati, S. Petit, S.M. Ray, D. Ham, C. Capers, H. Ewing, N. Coffin, L. C. McDonald, J. Jernigan, D. Cardo, Vital Signs: epidemiology and recent trends in methicillin-resistant and in methicillin-susceptible *Staphylococcus aureus* bloodstream infections — United States, *MMWR Morb. Mortal. Wkly. Rep.* 68 (2019) 214–219, <https://doi.org/10.15585/mmwr.mm6809e1>.
- [6] A.S. Lee, H. de Lencastre, J. Garau, J. Kluytmans, S. Malhotra-Kumar, A. Peschel, S. Harbarth, Methicillin-resistant *Staphylococcus aureus*, *Nat. Rev. Dis. Prim.* 4 (2018) 18033, <https://doi.org/10.1038/nrdp.2018.33>.
- [7] P. Nandhini, P. Kumar, S. Mickymaray, A.S. Alothaim, J. Somasundaram, M. Rajan, Recent developments in methicillin-resistant *Staphylococcus aureus* (MRSA) treatment: a review, *Antibiotics* 11 (2022) 606, <https://doi.org/10.3390/antibiotics11050606>.
- [8] A. Sakr, F. Brégeon, J.-L. Mège, J.-M. Rolain, O. Blin, *Staphylococcus aureus* nasal colonization: an update on mechanisms, epidemiology, risk factors, and subsequent infections, *Front. Microbiol.* 9 (2018), <https://doi.org/10.3389/fmicb.2018.02419>.
- [9] B.P. Howden, S.G. Giulieri, T. Wong Fok Lung, S.L. Baines, L.K. Sharkey, J.Y.H. Lee, A. Hachani, I.R. Monk, T.P. Stinear, *Staphylococcus aureus* host interactions and adaptation, *Nat. Rev. Microbiol.* 21 (2023) 380–395, <https://doi.org/10.1038/s41579-023-00852-y>.
- [10] J.C. Lam, W. Stokes, The golden grapes of wrath – *Staphylococcus aureus* bacteremia: a clinical review, *Am. J. Med.* 136 (2023) 19–26, <https://doi.org/10.1016/j.amjmed.2022.09.017>.
- [11] L. Liesenborghs, S. Meyers, M. Lox, M. Criel, J. Claes, M. Peetermans, S. Trenson, G. Vande Velde, P. Vanden Bergh, P. Baatsen, D. Missiakos, O. Schneewind, W. E. Peetermans, M.F. Hoylaerts, T. Vanassche, P. Verhamme, *Staphylococcus aureus* endocarditis: distinct mechanisms of bacterial adhesion to damaged and inflamed heart valves, *Eur. Heart J.* 40 (2019) 3248–3259, <https://doi.org/10.1093/eurheartj/ehz175>.
- [12] M.S. Linz, A. Mattappallil, D. Finkel, D. Parker, Clinical impact of *Staphylococcus aureus* skin and soft tissue infections, *Antibiotics* 12 (2023) 557, <https://doi.org/10.3390/antibiotics12030557>.
- [13] K.L. Urish, J.E. Cassat, *Staphylococcus aureus* osteomyelitis: bone, bugs, and surgery, *Infect. Immun.* 88 (2020), <https://doi.org/10.1128/IAI.00932-19>.
- [14] T. Jin, M. Mohammad, R. Pullerits, A. Ali, Bacteria and host interplay in *Staphylococcus aureus* septic arthritis and sepsis, *Pathogens* 10 (2021) 158, <https://doi.org/10.3390/pathogens10020158>.
- [15] G. Pietrocola, D. Campoccia, C. Motta, L. Montanaro, C.R. Arciola, P. Speziale, Colonization and infection of indwelling medical devices by *Staphylococcus aureus* with an emphasis on orthopedic implants, *Int. J. Mol. Sci.* 23 (2022) 5958, <https://doi.org/10.3390/ijms23115958>.
- [16] E.J.M. Raineri, D. Altulea, J.M. van Dijk, *Staphylococcal trafficking and infection—from 'nose to gut' and back*, *FEMS Microbiol. Rev.* 46 (2022), <https://doi.org/10.1093/femsre/fuab041>.
- [17] H. Chen, J. Zhang, Y. He, Z. Lv, Z. Liang, J. Chen, P. Li, J. Liu, H. Yang, A. Tao, X. Liu, Exploring the role of *Staphylococcus aureus* in inflammatory diseases, *Toxins (Basel)* 14 (2022) 464, <https://doi.org/10.3390/toxins14070464>.
- [18] V. Pintado, R. Pazos, M.E. Jiménez-Mejías, A. Rodríguez-Guardado, B. Díaz-Pollán, C. Cabellos, J.M. García-Lechuz, J. Lora-Tamayo, P. Domingo, E. Muñoz, D. Domingo, F. González-Romo, J.A. Lepe-Jiménez, C. Rodríguez-Lucas, A. Gil, I. Pelegrín, F. Chaves, V. Pomar, A. Ramos, T. Alarcón, E. Pérez-Cecilia, *Staphylococcus aureus* meningitis in adults: a comparative cohort study of infections caused by methicillin-resistant and methicillin-susceptible strains, *J. Hosp. Infect.* 102 (2019) 108–115, <https://doi.org/10.1016/j.jhin.2018.11.008>.
- [19] B. Poudel, Q. Zhang, A. Trongtorsak, B. Pyakuryal, G. Egorian, M. Sous, R. Ahmed, D.P. Trelles-García, M.A. Yanez-Bello, V.P. Trelles-García, J.J. Stake, G. Rodríguez-Nava, An overlooked cause of septic shock: staphylococcal toxic shock syndrome secondary to an axillary abscess, *Idcases* 23 (2021) e01039.
- [20] K. Xu, Y. Wang, Y. Jian, T. Chen, Q. Liu, H. Wang, M. Li, L. He, *Staphylococcus aureus* ST1 promotes persistent urinary tract infection by highly expressing the urease, *Front. Microbiol.* 14 (2023), <https://doi.org/10.3389/fmicb.2023.1101754>.
- [21] C.J. Heaton, G.R. Gerbig, L.D. Sensius, V. Patel, T.C. Smith, *Staphylococcus aureus* epidemiology in wildlife: a systematic review, *Antibiotics* 9 (2020) 89, <https://doi.org/10.3390/antibiotics9020089>.
- [22] E.Y. Garoy, Y.B. Gebreab, O.O. Achila, D.G. Tekeste, R. Kesete, R. Ghirmay, R. Kiflay, T. Tesfu, Methicillin-resistant *Staphylococcus aureus* (MRSA): prevalence and antimicrobial sensitivity pattern among patients—a multicenter study in asmara, eritrea, *Can. J. Infect. Dis. Med. Microbiol.* 2019 (2019) 1–9, <https://doi.org/10.1155/2019/8321834>.
- [23] M.A. Salam, M.Y. Al-Amin, M.T. Salam, J.S. Pawar, N. Akhter, A.A. Rabaan, M.A. Alqumber, Antimicrobial resistance: a growing serious threat for global public health, *Healthcare* 11 (2023) 1946, <https://doi.org/10.3390/healthcare11131946>.
- [24] B. Maddiboyina, H. Roy, M. Ramaiah, C.N. Sarvesh, S.H. Kosuru, R.K. Nakkala, B. S. Nayak, Methicillin-resistant *Staphylococcus aureus*: novel treatment approach breakthroughs, *Bulletin of the Georgian Academy of Sciences Natl. Res. Cent.* 47 (2023) 95, <https://doi.org/10.1186/s42269-023-01072-3>.
- [25] Y.N.V. Sabino, P.D. Cotter, H.C. Mantovani, Anti-virulence compounds against *Staphylococcus aureus* associated with bovine mastitis: a new therapeutic option? *Microbiol. Res.* 271 (2023) 127345, <https://doi.org/10.1016/j.micres.2023.127345>.
- [26] D. Raafat, M. Otto, K. Reppschlager, J. Iqbal, S. Holtfreter, Fighting *Staphylococcus aureus* biofilms with monoclonal antibodies, *Trends Microbiol.* 27 (2019) 303–322, <https://doi.org/10.1016/j.tim.2018.12.009>.
- [27] I.P. Thomsen, G. Sapparapu, D.B.A. James, J.E. Cassat, M. Nagarsheth, N. Kose, N. Putnam, K.M. Boguslawski, L.S. Jones, J.B. Wood, C.B. Creech, V.J. Torres, J. E. Crowe, Monoclonal antibodies against the *Staphylococcus aureus* bicomponent leukotoxin AB isolated following invasive human infection reveal diverse binding and modes of action, *J. Infect. Dis.* 215 (2017) 1124–1131, <https://doi.org/10.1093/infdis/jix071>.
- [28] Y. Guo, G. Song, M. Sun, J. Wang, Y. Wang, Prevalence and therapies of antibiotic-resistance in *Staphylococcus aureus*, *Front. Cell. Infect. Microbiol.* 10 (2020), <https://doi.org/10.3389/fcimb.2020.00107>.
- [29] R.A. Elmesser, S.E. Saleh, H.M. Elsherif, I.S. Yahia, K.M. Aboshanab, Staphyloxanthin as a potential novel target for deciphering promising anti-*Staphylococcus aureus* agents, *Antibiotics* 11 (2022) 298, <https://doi.org/10.3390/antibiotics11030298>.
- [30] A.E. Campbell, A.R. McCready-Vangi, A. Uberoi, S.M. Murga-Garrido, V. M. Lovins, E.K. White, J.-T.-C. Pan, S.A.B. Knight, A.R. Morgenstern, C. Bianco, P. J. Planet, S.E. Gardner, E.A. Grice, Variable staphyloxanthin production by *Staphylococcus aureus* drives strain-dependent effects on diabetic wound-healing outcomes, *Cell Rep.* 42 (2023) 113281, <https://doi.org/10.1016/j.celrep.2023.113281>.
- [31] W. Paulander, A.N. Varming, M.S. Bojer, C. Friberg, K. Bæk, H. Ingmer, The *agr* quorum sensing system in *Staphylococcus aureus* cells mediates death of sub-population, *BMC Res. Notes* 11 (2018) 503, <https://doi.org/10.1186/s13104-018-3600-6>.
- [32] P. Piewngam, J. Chiou, P. Chatterjee, M. Otto, Alternative approaches to treat bacterial infections: targeting quorum-sensing, *Expert Rev. Anti. Infect. Ther.* 18 (2020) 499–510, <https://doi.org/10.1080/14787210.2020.1750951>.

- [33] T.J. Polaske, K.H.J. West, K. Zhao, D.L. Widner, J.T. York, H.E. Blackwell, Chemical and biomolecular insights into the staphylococcus aureus Agr quorum sensing system: current progress and ongoing challenges, *Isr. J. Chem.* 63 (2023), <https://doi.org/10.1002/ijch.202200096>.
- [34] B.F. Ricciardi, G. Muthukrishnan, E. Masters, M. Ninomiya, C.C. Lee, E. M. Schwarz, Staphylococcus aureus evasion of host immunity in the setting of prosthetic joint infection: biofilm and beyond, *Curr. Rev. Musculoskelet. Med.* 11 (2018) 389–400, <https://doi.org/10.1007/s12178-018-9501-4>.
- [35] O.F. Martínez, H.M. Duque, O.L. Franco, Peptidomimetics as potential anti-virulence drugs against resistant bacterial pathogens, *Front. Microbiol.* 13 (2022), <https://doi.org/10.3389/fmicb.2022.831037>.
- [36] S.Y.C. Tong, J.S. Davis, E. Eichenberger, T.L. Holland, V.G. Fowler, Staphylococcus aureus Infections: epidemiology, pathophysiology, clinical manifestations, and management, *Clin. Microbiol. Rev.* 28 (2015) 603–661, <https://doi.org/10.1128/CMR.00134-14>.
- [37] A.F. Brown, J.M. Leech, T.R. Rogers, R.M. McLoughlin, Staphylococcus aureus colonization: modulation of host immune response and impact on human vaccine design, *Front. Immunol.* 4 (2014), <https://doi.org/10.3389/fimmu.2013.00507>.
- [38] A.-M. Hanssen, B. Kindlund, N.C. Stenklev, A.-S. Furberg, S. Fismen, R.S. Olsen, M. Johannessen, J.U.E. Sollid, Localization of Staphylococcus aureus in tissue from the nasal vestibule in healthy carriers, *BMC Microbiol.* 17 (2017) 89, <https://doi.org/10.1186/s12866-017-0997-3>.
- [39] A.-K.-E. Gehrke, C. Gai, M.I. Gómez, Staphylococcus aureus adaptation to the skin in health and persistent/recurrent infections, *Antibiotics* 12 (2023) 1520, <https://doi.org/10.3390/antibiotics12101520>.
- [40] D. Crespo-Piauelo, P.G. Lawlor, Livestock-associated methicillin-resistant Staphylococcus aureus (LA-MRSA) prevalence in humans in close contact with animals and measures to reduce on-farm colonisation, *Ir. Vet. J.* 74 (2021) 21, <https://doi.org/10.1186/s13620-021-00200-7>.
- [41] G.R. Pidwill, J.F. Gibson, J. Cole, S.A. Renshaw, S.J. Foster, The role of macrophages in staphylococcus aureus infection, *Front. Immunol.* 11 (2021), <https://doi.org/10.3389/fimmu.2020.620339>.
- [42] K. Tam, V.J. Torres, Staphylococcus aureus secreted toxins and extracellular enzymes, *Microbiol. Spectr.* 7 (2019), <https://doi.org/10.1128/microbiolspec.GPP3-0039-2018>.
- [43] V. Singh, U.J. Phukan, Interaction of host and Staphylococcus aureus protease-system regulates virulence and pathogenicity, *Med. Microbiol. Immunol.* 208 (2019) 585–607, <https://doi.org/10.1007/s00430-018-0573-y>.
- [44] L.M. Popov, C.D. Marceau, P.M. Stark, J.H. Lumb, J. Shah, D. Guerrero, R. L. Cooper, C. Merakou, D.M. Bouley, W. Meng, H. Kiyonari, M. Takeichi, S. J. Galli, F. Bagnoli, S. Citi, J.E. Carette, M.R. Amieva, The adherens junctions control susceptibility to Staphylococcus aureus α -toxin, *Proc. Natl. Acad. Sci.* 112 (2015) 14337–14342, <https://doi.org/10.1073/pnas.1510265112>.
- [45] J. Bien, O. Sokolova, P. Bozko, Characterization of virulence factors of Staphylococcus aureus: novel function of known virulence factors that are implicated in activation of airway epithelial proinflammatory response, *J. Pathog.* 2011 (2011) 1–13, <https://doi.org/10.4061/2011/601905>.
- [46] Y. Zheng, L. He, T.K. Asiamah, M. Otto, Colonization of medical devices by staphylococci, *Environ. Microbiol.* 20 (2018) 3141–3153, <https://doi.org/10.1111/1462-2920.14129>.
- [47] P.M. Schlievert, C.C. Davis, Device-associated menstrual toxic shock syndrome, *Clin. Microbiol. Rev.* 33 (2020), <https://doi.org/10.1128/CMR.00032-19>.
- [48] N. Wunsch, S.J. Green, S. Adam, J. Hampton, P.A. Phillips-Howard, S.D. Mehta, In vitro study to assess effective cleaning techniques for removing Staphylococcus aureus from menstrual cups, *Int. J. Environ. Res. Public Health.* 19 (2022) 1450, <https://doi.org/10.3390/ijerph19031450>.
- [49] J. Kadariya, T.C. Smith, D. Thapaliya, Staphylococcus aureus and staphylococcal food-borne disease: an ongoing challenge in public health, *Biomed Res. Int.* 2014 (2014) 1–9, <https://doi.org/10.1155/2014/827965>.
- [50] H.K. Ono, S. Hirose, K. Narita, M. Sugiyama, K. Asano, D.-L. Hu, A. Nakane, Histamine release from intestinal mast cells induced by staphylococcal enterotoxin A (SEA) evokes vomiting reflex in common marmoset, *PLoS Pathog.* 15 (2019) e1007803.
- [51] K. Kulhankova, K.J. Kinney, J.M. Stach, F.A. Gourronc, I.M. Grumbach, A. J. Klingelhut, W. Salgado-Pabón, The superantigen toxic shock syndrome toxin 1 alters human aortic endothelial cell function, *Infect. Immun.* 86 (2018), <https://doi.org/10.1128/IAI.00848-17>.
- [52] L.M. Palma Medina, A.-K. Becker, S. Michalik, K. Surmann, P. Hildebrandt, M. Gesell Salazar, S.A. Mekonnen, L. Kaderali, U. Völker, J.M. van Dijk, Interaction of Staphylococcus aureus and host cells upon infection of bronchial epithelium during different stages of regeneration, *ACS Infect. Dis.* 6 (2020) 2279–2290, <https://doi.org/10.1021/acsfeddis.0c00403>.
- [53] H. Liu, N.K. Archer, C.A. Dillen, Y. Wang, A.G. Ashbaugh, R.V. Ortines, T. Kao, S. K. Lee, S.S. Cai, R.J. Miller, M.C. Marchitto, E. Zhang, D.P. Riggins, R.D. Plaut, S. Stibitz, R.S. Geha, L.S. Miller, Staphylococcus aureus epicutaneous exposure drives skin inflammation via IL-36-mediated T cell responses, *Cell Host Microbe.* 22 (2017) 653–666.e5, <https://doi.org/10.1016/j.chom.2017.10.006>.
- [54] H. Yamada, A. Kaitani, K. Izawa, T. Ando, A. Kamei, S. Uchida, A. Maehara, M. Kojima, R. Yamamoto, H. Wang, M. Nagamine, K. Maeda, K. Uchida, N. Nakano, Y. Ohtsuka, H. Ogawa, K. Okumura, T. Shimizu, J. Kitauro, Staphylococcus aureus δ -toxin present on skin promotes the development of food allergy in a murine model, *Front. Immunol.* 14 (2023), <https://doi.org/10.3389/fimmu.2023.1173069>.
- [55] K. Reshamwala, G.Y.C. Cheung, R.C. Hsieh, R. Liu, H.-S. Joo, Y. Zheng, J.S. Bae, T.H. Nguyen, A.E. Villaruz, A.S. Gozalo, W.R. Elkins, M. Otto, Identification and characterization of the pathogenic potential of phenol-soluble modulin toxins in the mouse commensal Staphylococcus xyloso, *Front. Immunol.* 13 (2022), <https://doi.org/10.3389/fimmu.2022.999201>.
- [56] S.Y. Aboagye, G. Kpeli, J. Tuffour, D. Yeboah-Manu, Challenges associated with the treatment of Buruli ulcer, *J. Leukoc. Biol.* 105 (2019) 233–242, <https://doi.org/10.1002/JLB.MR0318-128>.
- [57] S. Tsuzuki, N. Matsunaga, K. Yahara, Y. Gu, K. Hayakawa, A. Hirabayashi, T. Kajihara, M. Sugai, K. Shibayama, N. Ohmagari, National trend of blood-stream infection attributable deaths caused by Staphylococcus aureus and Escherichia coli in Japan, *J. Infect. Chemother.* 26 (2020) 367–371, <https://doi.org/10.1016/j.jiac.2019.10.017>.
- [58] D. Chinemerem Nwobodo, M.C. Ugwu, C. Oliseloke Anie, M.T.S. Al-Ouqaili, J. Chinedu Ikem, U. Victor Chigozie, M. Saki, Antibiotic resistance: the challenges and some emerging strategies for tackling a global menace, *J. Clin. Lab. Anal.* 36 (2022), <https://doi.org/10.1002/jcla.24655>.
- [59] B. Ali Alghamdi, I. Al-Johani, J.M. Al-Shamrani, H. Musamed Alshamrani, B. G. Al-Otaibi, K. Almazmomi, N. Yusnoraini Yusuf, Antimicrobial resistance in methicillin-resistant staphylococcus aureus, Saudi, *J. Biol. Sci.* 30 (2023) 103604, <https://doi.org/10.1016/j.sjbs.2023.103604>.
- [60] K.I.K.N. Kochhar, A. Ghosh, S. Shrivastava, V. Singh Rawat, S. Mondal Ghorai, K. Kaur Sodhi, A. James, M. Kumar, Perspectives on systematic generation of antibiotic resistance with special emphasis on modern antibiotics, *Total Environ. Res. Themes.* 8 (2023) 100068, <https://doi.org/10.1016/j.totres.2023.100068>.
- [61] B. Mlynarczyk-Bonikowska, C. Kowalewski, A. Krolak-Ulinska, W. Marusza, Molecular mechanisms of drug resistance in Staphylococcus aureus, *Int. J. Mol. Sci.* 23 (2022) 8088, <https://doi.org/10.3390/ijms23158088>.
- [62] H. Lade, J.-S. Kim, Molecular determinants of β -lactam resistance in methicillin-resistant Staphylococcus aureus (MRSA): an updated review, *Antibiotics* 12 (2023) 1362, <https://doi.org/10.3390/antibiotics12091362>.
- [63] H. Lade, J.-S. Kim, Bacterial targets of antibiotics in methicillin-resistant Staphylococcus aureus, *Antibiotics* 10 (2021) 398, <https://doi.org/10.3390/antibiotics10040398>.
- [64] M. Gopikrishnan, S. Haryini, Emerging strategies and therapeutic innovations for combating drug resistance in Staphylococcus aureus strains: a comprehensive review, *J. Basic Microbiol.* (2024), <https://doi.org/10.1002/jobm.202300579>.
- [65] H. Lade, H.-S. Joo, J.-S. Kim, Molecular basis of non- β -Lactam antibiotics resistance in Staphylococcus aureus, *Antibiotics* 11 (2022) 1378, <https://doi.org/10.3390/antibiotics11101378>.
- [66] O.E. Akanbi, H.A. Njom, J. Fri, A.C. Otigbu, A.M. Clarke, Antimicrobial susceptibility of Staphylococcus aureus isolated from recreational waters and beach sand in Eastern Cape Province of South Africa, *Int. J. Environ. Res. Public Health.* 14 (2017) 1001, <https://doi.org/10.3390/ijerph14091001>.
- [67] S. Grillo, M. Puig-Asensio, M.L. Schweizer, G. Cuervo, I. Oriol, M. Pujol, J. Carratalá, The Effectiveness of combination therapy for treating methicillin-susceptible Staphylococcus aureus bacteremia: a systematic literature review and a meta-analysis, *Microorganisms* 10 (2022) 848, <https://doi.org/10.3390/microorganisms10050848>.
- [68] W. Rose, M. Fantl, M. Geriak, V. Nizet, G. Sakoulas, Current paradigms of combination therapy in methicillin-resistant Staphylococcus aureus (MRSA) bacteremia: does it work, which combination, and for which patients? *Clin. Infect. Dis.* 73 (2021) 2353–2360, <https://doi.org/10.1093/cid/ciab452>.
- [69] M. Wang, Y. Lian, Y. Wang, L. Zhu, The role and mechanism of quorum sensing on environmental antimicrobial resistance, *Environ. Pollut.* 322 (2023) 121238, <https://doi.org/10.1016/j.envpol.2023.121238>.
- [70] K. Vadakkan, A.A. Choudhury, R. Gunasekaran, J. Hemapriya, S. Vijayanand, Quorum sensing intervened bacterial signaling: pursuit of its cognizance and repression, *J. Genet. Eng. Biotechnol.* 16 (2018) 239–252, <https://doi.org/10.1016/j.jgeb.2018.07.001>.
- [71] Q. Peng, X. Tang, W. Dong, N. Sun, W. Yuan, A review of biofilm formation of Staphylococcus aureus and its regulation mechanism, *Antibiotics* 12 (2022) 12, <https://doi.org/10.3390/antibiotics12010012>.
- [72] L. Tan, S.R. Li, B. Jiang, X.M. Hu, S. Li, Therapeutic targeting of the staphylococcus aureus accessory gene regulator (agr) system, *Front. Microbiol.* 9 (2018), <https://doi.org/10.3389/fmicb.2018.00055>.
- [73] Y. Wang, B. Liu, D. Grenier, L. Yi, Regulatory mechanisms of the LuxS/AI-2 system and bacterial resistance, *Antimicrob. Agents Chemother.* 63 (2019), <https://doi.org/10.1128/AAC.01186-19>.
- [74] D. Joshi, C. Chbib, M.N. Uddin, M.J. D'Souza, Evaluation of microparticulate (S)-4,5-dihydroxy-2,3-pentanedione (DPD) as a potential vaccine adjuvant, *AAPS J.* 23 (2021) 84, <https://doi.org/10.1208/s12248-021-00617-6>.
- [75] Y. Li, T. Feng, Y. Wang, The role of bacterial signaling networks in antibiotics response and resistance regulation, *Mar. Life Sci. Technol.* 4 (2022) 163–178, <https://doi.org/10.1007/s42995-022-00126-1>.
- [76] Y. Wang, B. Liu, J. Li, S. Gong, X. Dong, C. Mao, L. Yi, LuxS/AI-2 system is involved in fluoroquinolones susceptibility in Streptococcus suis through overexpression of efflux pump SatAB, *Vet. Microbiol.* 233 (2019) 154–158, <https://doi.org/10.1016/j.vetmic.2019.05.006>.
- [77] V. Vinodhini, M. Kavitha, Deciphering agr quorum sensing in Staphylococcus aureus: insights and therapeutic prospects, *Mol. Biol. Rep.* 51 (2024) 155, <https://doi.org/10.1007/s11033-023-08930-3>.
- [78] H.T.T. Nguyen, T.H. Nguyen, M. Otto, The staphylococcal exopolysaccharide PIA – Biosynthesis and role in biofilm formation, colonization, and infection, *Comput. Struct. Biotechnol. J.* 18 (2020) 3324–3334, <https://doi.org/10.1016/j.csbj.2020.10.027>.
- [79] Y. Mao, P. Liu, H. Chen, Y. Wang, C. Li, Q. Wang, Baicalein inhibits the Staphylococcus aureus biofilm and the LuxS/AI-2 system in vitro, *Infect. Drug Resist.* 16 (2023) 2861–2882, <https://doi.org/10.2147/IDR.S406243>.

- [80] G. Grassia, J. Bagnarino, M. Siciliano, D. Barbarini, M. Corbella, P. Cambieri, F. Baldanti, V. Monzillo, Phenotypic and genotypic assays to evaluate coagulase-negative staphylococci biofilm production in bloodstream infections, *Microorganisms* 12 (2024) 126, <https://doi.org/10.3390/microorganisms12010126>.
- [81] Z. Jin, Q. Jiang, B. Fang, B. Sun, The ArlR-MgrA regulatory cascade regulates PIA-dependent and protein-mediated biofilm formation in Rbf-dependent and Rbf-independent pathways, *Int. J. Med. Microbiol.* 309 (2019) 85–96, <https://doi.org/10.1016/j.ijmm.2018.12.006>.
- [82] D. Francis, G. Veeramani, Kathadathil Hari, A. Koonthanma Subash, A. Bhairaddy, A. Joy, The biofilm proteome of *Staphylococcus aureus* and its implications for therapeutic interventions to biofilm-associated infections, in: 2024: pp. 327–400, <https://doi.org/10.1016/bs.apcsb.2023.08.002>.
- [83] B. Wang, T.W. Muir, regulation of virulence in *Staphylococcus aureus*: molecular mechanisms and remaining puzzles, *Cell Chem. Biol.* 23 (2016) 214–224, <https://doi.org/10.1016/j.chembiol.2016.01.004>.
- [84] L. Tan, Y. Huang, W. Shang, Y. Yang, H. Peng, Z. Hu, Y. Wang, Y. Rao, Q. Hu, X. Rao, X. Hu, M. Li, K. Chen, S. Li, Accessory gene regulator (agr) allelic variants in cognate *Staphylococcus aureus* strain display similar phenotypes, *Front. Microbiol.* 13 (2022), <https://doi.org/10.3389/fmicb.2022.700894>.
- [85] M. Grazul, E. Balcerzak, M. Sienkiewicz, Analysis of the Presence of the virulence and regulation genes from *Staphylococcus aureus* (S. aureus) in coagulase negative staphylococci and the influence of the staphylococcal cross-talk on their functions, *Int. J. Environ. Res. Public Health* 20 (2023) 5155, <https://doi.org/10.3390/ijerph20065155>.
- [86] S.K. Srivastava, K. Rajasree, A. Fasim, G. Arakere, B. Gopal, Influence of the AgrC-AgrA complex on the response time of *Staphylococcus aureus* quorum sensing, *J. Bacteriol.* 196 (2014) 2876–2888, <https://doi.org/10.1128/JB.01530-14>.
- [87] A. Zhao, S.P. Bodine, Q. Xie, B. Wang, G. Ram, R.P. Novick, T.W. Muir, Reconstitution of the S. aureus agr quorum sensing pathway reveals a direct role for the integral membrane protease MroQ in pheromone biosynthesis, *Proc. Natl. Acad. Sci.* 119 (2022), <https://doi.org/10.1073/pnas.2202661119>.
- [88] P. Williams, P. Hill, B. Bonev, W.C. Chan, Quorum-sensing, intra- and inter-species competition in the staphylococci, *Microbiology* 169 (2023), <https://doi.org/10.1099/mic.0.001381>.
- [89] E.L. Fisher, M. Otto, G.Y.C. Cheung, Basis of virulence in enterotoxin-mediated staphylococcal food poisoning, *Front. Microbiol.* 9 (2018), <https://doi.org/10.3389/fmicb.2018.00436>.
- [90] S. Jarraud, G.J. Lyon, A.M.S. Figueiredo, G. Lina, F. Vandenesch, J. Etienne, T. W. Muir, R.P. Novick, Exfoliating-producing strains define a fourth agr specificity group in *Staphylococcus aureus*, *J. Bacteriol.* 193 (2011) 7027, <https://doi.org/10.1128/JB.06355-11>.
- [91] J.K. Vasquez, H.E. Blackwell, Simplified autoinducing peptide mimetics with single-nanomolar activity against the *Staphylococcus aureus* AgrC quorum sensing receptor, *ACS Infect. Dis.* 5 (2019) 484–492, <https://doi.org/10.1021/acscinfed.9b00002>.
- [92] M.R. Stock, L. Fang, K.R. Johnson, C. Cosgriff, W.P. Teoh, F. Alonzo, Characterization of MroQ-dependent maturation and export of the *Staphylococcus aureus* accessory gene regulatory system autoinducing peptide, *Infect. Immun.* 90 (2022), <https://doi.org/10.1128/iai.00263-22>.
- [93] P. Bardelang, E.J. Murray, I. Blower, S. Zandomenighi, A. Goode, R. Hussain, D. Kumari, G. Siligardi, K. Inoue, J. Luckett, J. Douth, J. Emsley, W.C. Chan, P. Hill, P. Williams, B.B. Bonev, Conformational analysis and interaction of the *Staphylococcus aureus* transmembrane peptidase AgrB with its AgrD propeptide substrate, *Front. Chem.* 11 (2023), <https://doi.org/10.3389/fchem.2023.1113885>.
- [94] C. Jenul, A.R. Horswill, Regulation of *Staphylococcus aureus* virulence, *Microbiol. Spectr.* 7 (2019), <https://doi.org/10.1128/microbiolspec.GPP3-0031-2018>.
- [95] F.F. Tuon, P.H. Suss, J.P. Telles, L.R. Dantas, N.H. Borges, V.S.T. Ribeiro, Antimicrobial treatment of *Staphylococcus aureus* biofilms, *Antibiotics* 12 (2023) 87, <https://doi.org/10.3390/antibiotics12010087>.
- [96] S. Azimi, A.D. Klementiev, M. Whiteley, S.P. Diggle, Bacterial quorum sensing during infection, *Annu. Rev. Microbiol.* 74 (2020) 201–219, <https://doi.org/10.1146/annurev-micro-032020-093845>.
- [97] N. Ahmad-Mansour, P. Loubet, C. Pouget, C. Donyach-Remy, A. Sotto, J.-P. Lavigne, V. Molle, *Staphylococcus aureus* toxins: an update on their pathogenic properties and potential treatments, *Toxins (Basel)* 13 (2021) 677, <https://doi.org/10.3390/toxins13100677>.
- [98] A. Zhao, J. Sun, Y. Liu, Understanding bacterial biofilms: from definition to treatment strategies, *Front. Cell. Infect. Microbiol.* 13 (2023), <https://doi.org/10.3389/fcimb.2023.1137947>.
- [99] C. Oriol, L. Cengher, A.C. Manna, T. Mauro, M.-L. Pinel-Marie, B. Felden, A. Cheung, A. Rouillon, Expanding the *Staphylococcus aureus* SarA regulon to small RNAs, *Msystems* 6 (2021), <https://doi.org/10.1128/mSystems.00713-21>.
- [100] A. Selvaraj, T. Jayasree, A. Valliammai, S.K. Pandian, Myrtenol attenuates MRSA biofilm and virulence by suppressing sarA expression dynamism, *Front. Microbiol.* 10 (2019), <https://doi.org/10.3389/fmicb.2019.02027>.
- [101] H. Patel, S. Rawat, A genetic regulatory see-saw of biofilm and virulence in MRSA pathogenesis, *Front. Microbiol.* 14 (2023), <https://doi.org/10.3389/fmicb.2023.1204428>.
- [102] A.H. Corbett, Post-transcriptional regulation of gene expression and human disease, *Curr. Opin. Cell Biol.* 52 (2018) 96–104, <https://doi.org/10.1016/j.cob.2018.02.011>.
- [103] B. Kim, H.-J. Lee, S.-H. Jo, M.-G. Kim, Y. Lee, W. Lee, W. Kim, H.-S. Joo, Y.-G. Kim, J.-S. Kim, Y.-H. Yang, Study of SarA by DNA affinity capture assay (DACA) employing three promoters of key virulence and resistance genes in methicillin-resistant *Staphylococcus aureus*, *Antibiotics* 11 (2022) 1714, <https://doi.org/10.3390/antibiotics11121714>.
- [104] P. François, J. Schrenzel, F. Götz, Biology and regulation of staphylococcal biofilm, *Int. J. Mol. Sci.* 24 (2023) 5218, <https://doi.org/10.3390/ijms24065218>.
- [105] B. Liu, B. Sun, Rsp promotes the transcription of virulence factors in an agr-independent manner in *Staphylococcus aureus*, *Emerg. Microbes Infect.* 9 (2020) 796–812, <https://doi.org/10.1080/22221751.2020.1752116>.
- [106] R. Mishra Awdhesh Kumar Mishra, B. Mishra Awdhesh Kumar Mishra, N. Easwaran, K. Muthukaliannan Gothandam, *Staphylococcus aureus* and virulence-related small RNA, in: *Insights Into Drug Resist. Staphylococcus Aureus*, 2021: p. 252. <https://doi.org/10.5772/intechopen.98520>.
- [107] D. Reyes, D.O. Andrey, A. Monod, W.L. Kelley, G. Zhang, A.L. Cheung, Coordinated regulation by AgrA, SarA, and SarR To control agr expression in *Staphylococcus aureus*, *J. Bacteriol.* 193 (2011) 6020–6031, <https://doi.org/10.1128/JB.05436-11>.
- [108] C.P. Gordon, P. Williams, W.C. Chan, Attenuating *Staphylococcus aureus* virulence gene regulation: a medicinal chemistry perspective, *J. Med. Chem.* 56 (2013) 1389–1404, <https://doi.org/10.1021/jm3014635>.
- [109] Z. Guan, Y. Liu, C. Liu, H. Wang, J. Feng, G. Yang, *Staphylococcus aureus* β -hemolysin up-regulates the expression of IFN- γ by human CD56bright NK cells, *Front. Cell. Infect. Microbiol.* 11 (2021), <https://doi.org/10.3389/fcimb.2021.658141>.
- [110] C.E. Butrico, J.E. Cassat, Quorum sensing and toxin production in *Staphylococcus aureus* osteomyelitis: pathogenesis and paradox, *Toxins (basel)* 12 (2020) 516, <https://doi.org/10.3390/toxins12080516>.
- [111] R.V. Sionov, D. Steinberg, Targeting the holy triangle of quorum sensing biofilm formation, and antibiotic resistance in pathogenic bacteria, *Microorganisms* 10 (2022) 1239, <https://doi.org/10.3390/microorganisms10061239>.
- [112] P. Speziale, G. Pietrocola, The multivalent role of fibronectin-binding proteins A and B (FnBPA and FnBPB) of *Staphylococcus aureus* in host infections, *Front. Microbiol.* 11 (2020), <https://doi.org/10.3389/fmicb.2020.02054>.
- [113] H. Oknin, Y. Kroupitski, M. Shemesh, S. Blum, Upregulation of ica operon governs biofilm formation by a coagulase-negative *Staphylococcus caprae*, *Microorganisms* 11 (2023) 1533, <https://doi.org/10.3390/microorganisms11061533>.
- [114] A.E. Paharik, A.R. Horswill, The staphylococcal biofilm: adhesins, regulation, and host response, *Microbiol. Spectr.* 4 (2016), <https://doi.org/10.1128/microbiolspec.VMBF-0022-2015>.
- [115] I.F. Bezar, A.A. Mashruwala, J.M. Boyd, A.M. Stock, Drug-like fragments inhibit agr-mediated virulence expression in *Staphylococcus aureus*, *Sci. Rep.* 9 (2019) 6786, <https://doi.org/10.1038/s41598-019-42853-z>.
- [116] S. Sharma, J. Mohler, S.D. Mahajan, S.A. Schwartz, L. Bruggemann, R. Aalinkel, Microbial biofilm: a review on formation infection antibiotic resistance, control measures, and innovative treatment, *Microorganisms* 11 (2023) 1614, <https://doi.org/10.3390/microorganisms11061614>.
- [117] Y. Liu, J. Zhang, Y. Ji, Environmental factors modulate biofilm formation by *Staphylococcus aureus*, *Sci. Prog.* 103 (2020) 003685041989865, <https://doi.org/10.1177/0036850419898659>.
- [118] G. Pietrocola, G. Nobile, S. Rindi, P. Speziale, *Staphylococcus aureus* manipulates innate immunity through own and host-expressed proteases, *Front. Cell. Infect. Microbiol.* 7 (2017), <https://doi.org/10.3389/fcimb.2017.00166>.
- [119] S.Y. Feng, Y. Hauck, F. Morgene, R. Mohammedi, N. Mirouze, The complex regulation of competence in *Staphylococcus aureus* under microaerobic conditions, *Commun. Biol.* 6 (2023) 512, <https://doi.org/10.1038/s42003-023-04892-1>.
- [120] S.-Y. Kim, M. Kim, T.-J. Kim, Regulation of σ B-dependent biofilm formation in *Staphylococcus aureus* through strain-specific signaling induced by diosgenin, *Microorganisms* 11 (2023) 2376, <https://doi.org/10.3390/microorganisms11102376>.
- [121] S. Supa-amornkul, P. Mongkolsuk, P. Sumppun, P. Chaiyakunvat, W. Navaratnaisit, C. Jiarapinitnun, S. Chaturongakul, Alternative sigma factor B in bovine mastitis-causing *Staphylococcus aureus*: characterization of its role in biofilm formation, resistance to hydrogen peroxide stress, regulon members, *Front. Microbiol.* 10 (2019), <https://doi.org/10.3389/fmicb.2019.02493>.
- [122] D.O. Andrey, A. Jousset, M. Villanueva, A. Renzoni, A. Monod, C. Barras, N. Rodriguez, W.L. Kelley, Impact of the regulators SigB, Rot, SarA and sarS on the toxic shock Tst promoter and TSST-1 expression in *Staphylococcus aureus*, *PLoS One* 10 (2015) e0135579.
- [123] A.K. Farha, Q.-Q. Yang, G. Kim, D. Zhang, V. Mavumengwana, O. Habimana, H.-B. Li, H. Corke, R.-Y. Gan, Inhibition of multidrug-resistant foodborne *Staphylococcus aureus* biofilms by a natural terpenoid (+)-nootkatone and related molecular mechanism, *Food Control.* 112 (2020) 107154, <https://doi.org/10.1016/j.foodcont.2020.107154>.
- [124] K. Vadakkan, S. Joy, V. Selvaraj, M.K. Cheruvathur, P. Remadevi, Biogenic nanoparticle-mediated inhibition of bacterial colonization and analyzing the effects of variables by response surface modeling, *Kuwait J. Sci.* 51 (2024) 100201, <https://doi.org/10.1016/j.kjs.2024.100201>.
- [125] C. Cheng, T. Jiang, D. Zhang, H. Wang, T. Fang, C. Li, Attachment characteristics and kinetics of biofilm formation by *Staphylococcus aureus* on ready-to-eat cooked beef contact surfaces, *J. Food Sci.* 88 (2023) 2595–2610, <https://doi.org/10.1111/1750-3841.16592>.
- [126] R. Konduri, C.R. Saiabhillash, S. Shivaji, Biofilm-forming potential of ocular fluid *Staphylococcus aureus* and *Staphylococcus epidermidis* on Ex Vivo human corneas from attachment to dispersal phase, *Microorganisms* 9 (2021) 1124, <https://doi.org/10.3390/microorganisms9061124>.

- [127] C. Burel, R. Dreyfus, L. Purevdorj-Gage, Physical mechanisms driving the reversible aggregation of *Staphylococcus aureus* and response to antimicrobials, *Sci. Rep.* 11 (2021) 15048, <https://doi.org/10.1038/s41598-021-94457-1>.
- [128] E. Maikranz, C. Spengler, N. Thewes, A. Thewes, F. Nolle, P. Jung, M. Bischoff, L. Santen, K. Jacobs, Different binding mechanisms of *Staphylococcus aureus* to hydrophobic and hydrophilic surfaces, *Nanoscale* 12 (2020) 19267–19275, <https://doi.org/10.1039/D0NR03134H>.
- [129] Y. Liu, J. Yang, M. Ji, J. Phillips, M. Wylam, Y. Ji, Identification of *cbiO* gene critical for biofilm formation by MRSA CFSa36 strain isolated from pediatric patient with cystic fibrosis, *Pathogens* 10 (2021) 1363, <https://doi.org/10.3390/pathogens10111363>.
- [130] S. Wu, Y. Liu, L. Lei, H. Zhang, An antisense *yycF* RNA modulates biofilm organization of methicillin-resistant *Staphylococcus aureus* and pathogenicity in a rat model of osteomyelitis, *Antibiotics* 10 (2021) 603, <https://doi.org/10.3390/antibiotics10050603>.
- [131] C. Pouget, C. Dunyach-Remy, C. Magnan, A. Pantel, A. Sotto, J.-P. Lavigne, Polymicrobial biofilm organization of *Staphylococcus aureus* and *Pseudomonas aeruginosa* in a chronic wound environment, *Int. J. Mol. Sci.* 23 (2022) 10761, <https://doi.org/10.3390/ijms231810761>.
- [132] Z.-J. Pei, C. Li, W. Dai, Z. Lou, X. Sun, H. Wang, A.A. Khan, C. Wan, The anti-biofilm activity and mechanism of apigenin-7-O-glucoside against *Staphylococcus aureus* and *Escherichia coli*, *Infect. Drug Resist.* 16 (2023) 2129–2140, <https://doi.org/10.2147/IDR.S387157>.
- [133] Y. Xia, P.G. Jayatilake, B. Li, P. Zuliani, D. Deehan, J. Longyear, P. Stoodley, J. Chen, Coupled CFD-DEM modeling to predict how EPS affects bacterial biofilm deformation, recovery and detachment under flow conditions, *Biotechnol. Bioeng.* 119 (2022) 2551–2563, <https://doi.org/10.1002/bit.28146>.
- [134] S. Yu, B. Jiang, C. Jia, H. Wu, J. Shen, X. Hu, Z. Xie, Investigation of biofilm production and its association with genetic and phenotypic characteristics of OM (osteomyelitis) and non-OM orthopedic *Staphylococcus aureus*, *Ann. Clin. Microbiol. Antimicrob.* 19 (2020) 10, <https://doi.org/10.1186/s12941-020-00352-4>.
- [135] M.H. Muhammad, A.L. Idris, X. Fan, Y. Guo, Y. Yu, X. Jin, J. Qiu, X. Guan, T. Huang, Beyond risk: bacterial biofilms and their regulating approaches, *Front. Microbiol.* 11 (2020), <https://doi.org/10.3389/fmicb.2020.00928>.
- [136] A. Herrera, B.G. Vu, C.S. Stach, J.A. Merriman, A.R. Horswill, W. Salgado-Pabón, P.M. Schlievert, *Staphylococcus aureus* β -Toxin mutants are defective in biofilm ligase and sphingomyelinase activity, and causation of infective endocarditis and sepsis, *Biochemistry* 55 (2016) 2510–2517, <https://doi.org/10.1021/acs.biochem.6b00083>.
- [137] R. Roy, M. Tiwari, G. Donelli, V. Tiwari, Strategies for combating bacterial biofilms: a focus on anti-biofilm agents and their mechanisms of action, *Virulence* 9 (2018) 522–554, <https://doi.org/10.1080/21505594.2017.1313372>.
- [138] H. Guo, Y. Tong, J. Cheng, Z. Abbas, Z. Li, J. Wang, Y. Zhou, D. Si, R. Zhang, Biofilm and small colony variants—an update on *Staphylococcus aureus* strategies toward drug resistance, *Int. J. Mol. Sci.* 23 (2022) 1241, <https://doi.org/10.3390/ijms23031241>.
- [139] O. Ciofu, C. Moser, P.Ø. Jensen, N. Høiby, Tolerance and resistance of microbial biofilms, *Nat. Rev. Microbiol.* 20 (2022) 621–635, <https://doi.org/10.1038/s41579-022-00682-4>.
- [140] P. Arumugam, T. Kielian, Metabolism shapes immune responses to *Staphylococcus aureus*, *J. Innate Immun.* (2024) 12–30, <https://doi.org/10.1159/000535482>.
- [141] F. Peyrussion, T.K. Nguyen, T. Najdovski, F. Van Bambeke, Host cell oxidative stress induces dormant *Staphylococcus aureus* persisters, *Microbiol. Spectr.* 10 (2022), <https://doi.org/10.1128/spectrum.02313-21>.
- [142] J.W. Hart, T.A. Waigh, J.R. Lu, I.S. Roberts, Micro rheology and spatial heterogeneity of *Staphylococcus aureus* biofilms modulated by hydrodynamic shear and biofilm-degrading enzymes, *Langmuir* 35 (2019) 3553–3561, <https://doi.org/10.1021/acs.langmuir.8b04252>.
- [143] S. Singh, S. Datta, K.B. Narayanan, K.N. Rajnish, Bacterial exo-polysaccharides in biofilms: role in antimicrobial resistance and treatments, *J. Genet. Eng. Biotechnol.* 19 (2021) 140, <https://doi.org/10.1186/s43141-021-00242-y>.
- [144] L.A. Onyango, M.M. Alreshidi, Adaptive metabolism in staphylococci: survival and persistence in environmental and clinical settings, *J. Pathog.* 2018 (2018) 1–11, <https://doi.org/10.1155/2018/1092632>.
- [145] V. Silva, L. Almeida, V. Gaio, N. Cerca, V. Manageiro, M. Caniça, J.L. Capelo, G. Igrejas, P. Poeta, Biofilm formation of multidrug-resistant MRSA strains isolated from different types of human infections, *Pathogens* 10 (2021) 970, <https://doi.org/10.3390/pathogens10080970>.
- [146] E. Balducci, F. Papi, D.E. Capiabli, L. Del Bino, Polysaccharides' structures and functions in biofilm architecture of antimicrobial-resistant (AMR) pathogens, *Int. J. Mol. Sci.* 24 (2023) 4030, <https://doi.org/10.3390/ijms24044030>.
- [147] S. Sugimoto, F. Sato, R. Miyakawa, A. Chiba, S. Onodera, S. Hori, Y. Mizunoe, Broad impact of extracellular DNA on biofilm formation by clinically isolated Methicillin-resistant and -sensitive strains of *Staphylococcus aureus*, *Sci. Rep.* 8 (2018) 2254, <https://doi.org/10.1038/s41598-018-20485-z>.
- [148] D.E. Moormeier, K.W. Bayles, *Staphylococcus aureus* biofilm: a complex developmental organism, *Mol. Microbiol.* 104 (2017) 365–376, <https://doi.org/10.1111/mmi.13634>.
- [149] K. Drożdż, D. Ochońska, Ł. Ścibik, M. Goida-Cępa, K. Biegun, M. Brzywczy-Włoch, The frequency of occurrence of resistance and genes involved in the process of adhesion and accumulation of biofilm in *Staphylococcus aureus* strains isolated from tracheostomy tubes, *Microorganisms* 10 (2022) 1210, <https://doi.org/10.3390/microorganisms10061210>.
- [150] Z. Sedarat, A.W. Taylor-Robinson, Biofilm Formation by pathogenic bacteria: applying a *Staphylococcus aureus* model to appraise potential targets for therapeutic intervention, *Pathogens* 11 (2022) 388, <https://doi.org/10.3390/pathogens11040388>.
- [151] X. Wang, D.S. Uppu, S.W. Dickey, D.J. Burgin, M. Otto, J.C. Lee, *Staphylococcus aureus* delta toxin modulates both extracellular membrane vesicle biogenesis and amyloid formation, *MBio* 14 (2023), <https://doi.org/10.1128/mbio.01748-23>.
- [152] A. Damour, B. Robin, L. Deroche, L. Broutin, N. Bellin, J. Verdon, G. Lina, F. M. Leclère, M. Garcia, J. Crenniter, N. Lévêque, C. Bodet, Phenol-soluble modulins α are major virulence factors of *Staphylococcus aureus* secretome promoting inflammatory response in human epidermis, *Virulence* 12 (2021) 2474–2492, <https://doi.org/10.1080/21505594.2021.1975909>.
- [153] S. Periasamy, H.-S. Joo, A.C. Duong, T.-H.-L. Bach, V.Y. Tan, S.S. Chatterjee, G.Y. C. Cheung, M. Otto, How *Staphylococcus aureus* biofilms develop their characteristic structure, *Proc. Natl. Acad. Sci.* 109 (2012) 1281–1286, <https://doi.org/10.1073/pnas.1115006109>.
- [154] K.Y. Le, S. Dastgheyb, T.V. Ho, M. Otto, Molecular determinants of staphylococcal biofilm dispersal and structuring, *Front. Cell. Infect. Microbiol.* 4 (2014), <https://doi.org/10.3389/fcimb.2014.00167>.
- [155] R. Malishev, N. Salinas, J. Gibson, A.B. Eden, J. Mieres-Perez, Y.B. Ruiz-Blanco, O. Malka, S. Kolusheva, F.-G. Klärner, T. Schrader, E. Sanchez-Garcia, C. Wang, M. Landau, G. Bitan, R. Jelinek, Inhibition of *Staphylococcus aureus* biofilm-forming functional amyloid by molecular tweezers, *Cell Chem. Biol.* 28 (2021) 1310–1320.e5, <https://doi.org/10.1016/j.chembiol.2021.03.013>.
- [156] S. Chetri, The culmination of multidrug-resistant efflux pumps vs. meager antibiotic arsenal era: Urgent need for an improved new generation of EPIs, *Front. Microbiol.* 14 (2023), <https://doi.org/10.3389/fmicb.2023.1149418>.
- [157] X.-Z. Li, P. Plésiat, H. Nikaido, The challenge of efflux-mediated antibiotic resistance in gram-negative bacteria, *Clin. Microbiol. Rev.* 28 (2015) 337–418, <https://doi.org/10.1128/CMR.00117-14>.
- [158] B. Kaur, J. Gupta, S. Sharma, D. Sharma, S. Sharma, Focused review on dual inhibition of quorum sensing and efflux pumps: a potential way to combat multi drug resistant *Staphylococcus aureus* infections, *Int. J. Biol. Macromol.* 190 (2021) 33–43, <https://doi.org/10.1016/j.ijbiomac.2021.08.199>.
- [159] M.N. Hajiagha, H.S. Kafil, Efflux pumps and microbial biofilm formation, *Infect. Genet. Evol.* 112 (2023) 105459, <https://doi.org/10.1016/j.meegid.2023.105459>.
- [160] R.M. Fleeman, G. Debevec, K. Anttonen, J.L. Adams, R.G. Santos, G.S. Welmaker, R.A. Houghton, M.A. Giulianotti, L.N. Shaw, Identification of a novel polyamine scaffold with potent efflux pump inhibition activity toward multi-drug resistant bacterial pathogens, *Front. Microbiol.* 9 (2018), <https://doi.org/10.3389/fmicb.2018.01301>.
- [161] G. Harikumar, K. Krishnan, The growing menace of drug resistant pathogens and recent strategies to overcome drug resistance: a review, *J. King Saud Univ. - Sci.* 34 (2022) 101979, <https://doi.org/10.1016/j.jksus.2022.101979>.
- [162] A. Sharma, V. Gupta, R. Pathania, Efflux pump inhibitors for bacterial pathogens: from bench to bedside, *Indian J. Med. Res.* 149 (2019) 129, https://doi.org/10.4103/ijmr.IJMR_2079_17.
- [163] M. Cheesman, A. Ilanko, B. Blonk, I. Cock, Developing new antimicrobial therapies: are synergistic combinations of plant extracts/compounds with conventional antibiotics the solution? *Pharmacogn. Rev.* 11 (2017) 57, https://doi.org/10.4103/phrev.phrev_21_17.
- [164] S. Zimmermann, M. Klinger-Strobel, J.A. Bohnert, S. Wendler, J. Rödel, M. W. Pletz, B. Löffler, L. Tuchscher, Clinically approved drugs inhibit the *Staphylococcus aureus* multidrug NorA efflux pump and reduce biofilm formation, *Front. Microbiol.* 10 (2019), <https://doi.org/10.3389/fmicb.2019.02762>.
- [165] I. Alav, J.M. Sutton, K.M. Rahman, Role of bacterial efflux pumps in biofilm formation, *J. Antimicrob. Chemother.* 73 (2018) 2003–2020, <https://doi.org/10.1093/jac/dky042>.
- [166] S.S. Costa, B. Sobkowiak, R. Parreira, J.D. Edgeworth, M. Viveiros, T.G. Clark, I. Couto, Genetic diversity of *norA*, coding for a main efflux pump of *Staphylococcus aureus*, *Front. Genet.* 9 (2019), <https://doi.org/10.3389/fgene.2018.00710>.
- [167] P. Rezaie, M. Pourhajibagher, N. Chiniforush, N. Hosseini, A. Bahador, The effect of quorum-sensing and efflux pumps interactions in *Pseudomonas aeruginosa* against photooxidative stress, *J. Lasers Med. Sci.* 9 (2018) 161–167, <https://doi.org/10.15171/jlms.2018.30>.
- [168] K. Vadakkan, A.K. Nangbam, K. Sathishkumar, N.P. Rumjit, M.K. Cheruvathur, A review of chemical signaling pathways in the quorum sensing circuit of *Pseudomonas aeruginosa*, *Int. J. Biol. Macromol.* 254 (2024) 127861, <https://doi.org/10.1016/j.ijbiomac.2023.127861>.
- [169] P. Shree, C.K. Singh, K.K. Sodhi, J.N. Surya, D.K. Singh, Biofilms: Understanding the structure and contribution towards bacterial resistance in antibiotics, *Med. Microbiol.* 16 (2023) 100084, <https://doi.org/10.1016/j.medmic.2023.100084>.
- [170] R. Musiol, Efflux systems as a target for anti-biofilm nanoparticles: perspectives on emerging applications, *Expert Opin. Ther. Targets* 27 (2023) 953–963, <https://doi.org/10.1080/14728222.2023.2263910>.
- [171] M.F. Varela, J. Stephen, D. Bharti, M. Lekshmi, S. Kumar, Inhibition of multidrug efflux pumps belonging to the major facilitator superfamily in bacterial pathogens, *Biomedicines* 11 (2023) 1448, <https://doi.org/10.3390/biomedicines11051448>.
- [172] Y.A. Helmy, K. Taha-Abdelaziz, H.-A.-E.-H. Hawwas, S. Ghosh, S.S. AlKafaas, M. M.M. Moawad, E.M. Saied, I.I. Kassem, A.M.M. Mawad, Antimicrobial resistance and recent alternatives to antibiotics for the control of bacterial pathogens with

- an emphasis on foodborne pathogens, *Antibiotics* 12 (2023) 274, <https://doi.org/10.3390/antibiotics12020274>.
- [173] S. Nadar, T. Khan, S.G. Patching, A. Omri, Development of antibiofilm therapeutics strategies to overcome antimicrobial drug resistance, *Microorganisms* 10 (2022) 303, <https://doi.org/10.3390/microorganisms10020303>.
- [174] L. Zhou, Y. Zhang, Y. Ge, X. Zhu, J. Pan, Regulatory mechanisms and promising applications of quorum sensing-inhibiting agents in control of bacterial biofilm formation, *Front. Microbiol.* 11 (2020), <https://doi.org/10.3389/fmicb.2020.589640>.
- [175] A. Panjla, G. Kaul, A. Akhri, D. Saxena, S. Joshi, C. Modak, D. Kumari, A. Jain, S. Chopra, S. Verma, Targeting multidrug resistant *Staphylococcus aureus* with cationic chlorpromazine-peptide conjugates, *Chem. – an Asian J.* 18 (2023), <https://doi.org/10.1002/asia.202300169>.
- [176] S. Sabatini, M. Piccioni, T. Felicetti, S. De Marco, G. Manfroni, R. Paggiotti, M. Nocchetti, V. Cecchetti, D. Pietrella, Investigation on the effect of known potent *S. aureus* NorA efflux pump inhibitors on the staphylococcal biofilm formation, *RSC Adv.* 7 (2017) 37007–37014, <https://doi.org/10.1039/C7RA03859C>.
- [177] N. Chandan, R. Tambat, R. Kalia, G. Kumar, N. Mahey, S. Jachak, H. Nandanwar, Efflux pump inhibitory potential of indole derivatives as an arsenal against norA over-expressing *Staphylococcus aureus*, *Microbiol. Spectr.* 11 (2023), <https://doi.org/10.1128/spectrum.04876-22>.
- [178] C. Kourtesi, A.R. Ball, Y.-Y. Huang, S.M. Jachak, D.M.A. Vera, P. Khondkar, S. Gibbons, M.R. Hamblin, G.P. Tegos, Microbial efflux systems and inhibitors: approaches to drug discovery and the challenge of clinical implementation, *Open Microbiol. J.* 7 (2013) 34–52, <https://doi.org/10.2174/1874285801307010034>.
- [179] Y.C. Fiamegos, P.L. Kastiris, V. Exarchou, H. Han, A.M.J.J. Bonvin, J. Vervoort, K. Lewis, M.R. Hamblin, G.P. Tegos, Antimicrobial and efflux pump inhibitory activity of caffeoylquinic acids from *artemisia absinthium* against gram-positive pathogenic bacteria, *PLoS One* 6 (2011) e18127.
- [180] W. Luan, X. Liu, X. Wang, Y. An, Y. Wang, C. Wang, K. Shen, H. Xu, S. Li, M. Liu, L. Yu, Inhibition of drug resistance of *Staphylococcus aureus* by efflux pump inhibitor and autolysis inducer to strengthen the antibacterial activity of β -lactam drugs, *Polish J. Microbiol.* 68 (2019) 477–491, <https://doi.org/10.33073/pjm-2019-047>.
- [181] D. Parai, M. Banerjee, P. Dey, S.K. Mukherjee, Reserpine attenuates biofilm formation and virulence of *Staphylococcus aureus*, *Microb. Pathog.* 138 (2020) 103790, <https://doi.org/10.1016/j.micpath.2019.103790>.
- [182] E.E. Mgbeahuruik, M. Stålnacke, H. Vuorela, Y. Holm, Antimicrobial and synergistic effects of commercial piperine and piperlongumine in combination with conventional antimicrobials, *Antibiotics* 8 (2019) 55, <https://doi.org/10.3390/antibiotics8020055>.
- [183] B.C. Manifold-Wheeler, B.O. Elmore, K.D. Triplett, M.J. Castleman, M. Otto, P. R. Hall, Serum lipoproteins are critical for pulmonary innate defense against *Staphylococcus aureus* quorum sensing, *J. Immunol.* 196 (2016) 328–335, <https://doi.org/10.4049/jimmunol.1501835>.
- [184] J.W. Hommes, R.M. Kratofil, S. Wahlen, C.J.C. de Haas, R.B. Hildebrand, G. K. Hovingh, M. Otto, M. van Eck, M. Hoekstra, S.J.A. Korpelaar, B.G. J. Surewaard, High density lipoproteins mediate in vivo protection against staphylococcal phenol-soluble modulins, *Sci. Rep.* 11 (2021) 15357, <https://doi.org/10.1038/s41598-021-94651-1>.
- [185] A.D. Wilde, D.J. Snyder, N.E. Putnam, M.D. Valentino, N.D. Hammer, Z. R. Lonergan, S.A. Hinger, E.E. Aysanoa, C. Blanchard, P.M. Dunman, G. A. Wasserman, J. Chen, B. Shopsis, M.S. Gilmore, E.P. Skaar, J.E. Cassat, Bacterial hypoxic responses revealed as critical determinants of the host-pathogen outcome by TnSeq analysis of *Staphylococcus aureus* invasive infection, *PLoS Pathog.* 11 (2015) e1005341.
- [186] K. Arunachalam, P. Pandurangan, C. Shi, R. Lagoa, Regulation of *Staphylococcus aureus* virulence and application of nanotherapeutics to eradicate *S. aureus* infection, *Pharmaceutics* 15 (2023) 310, <https://doi.org/10.3390/pharmaceutics15020310>.
- [187] F. Sun, Q. Ji, M.B. Jones, X. Deng, H. Liang, B. Frank, J. Telser, S.N. Peterson, T. Bae, C. He, AirSR, a [2Fe-2S] cluster-containing two-component system, mediates global oxygen sensing and redox signaling in *Staphylococcus aureus*, *J. Am. Chem. Soc.* 134 (2012) 305–314, <https://doi.org/10.1021/ja2071835>.
- [188] K.D. Mlynek, W.E. Sause, D.E. Moormeier, M.R. Sadykov, K.R. Hill, V.J. Torres, K. W. Bayles, S.R. Brinsmade, Nutritional regulation of the *sae* two-component system by CodY in *Staphylococcus aureus*, *J. Bacteriol.* 200 (2018), <https://doi.org/10.1128/JB.00012-18>.
- [189] J.S. Kavanaugh, A.R. Horswill, Impact of environmental cues on staphylococcal quorum sensing and biofilm development, *J. Biol. Chem.* 291 (2016) 12556–12564, <https://doi.org/10.1074/jbc.R116.722710>.
- [190] D. Bronesky, Z. Wu, S. Marzi, P. Walter, T. Geissmann, K. Moreau, F. Vandenesch, I. Caldelari, P. Romby, *Staphylococcus aureus* RNAIII and its regulon link quorum sensing, stress responses, metabolic adaptation, and regulation of virulence gene expression, *Annu. Rev. Microbiol.* 70 (2016) 299–316, <https://doi.org/10.1146/annurev-micro-102215-095708>.
- [191] R. Arya, S.A. Princy, Exploration of modulated genetic circuits governing virulence determinants in *Staphylococcus aureus*, *Indian J. Microbiol.* 56 (2016) 19–27, <https://doi.org/10.1007/s12088-015-0555-3>.
- [192] A.M. Ramirez, K.E. Beenken, S.D. Byrum, A.J. Tackett, L.N. Shaw, B.D. Gimza, M. S. Smeltzer, SarA plays a predominant role in controlling the production of extracellular proteases in the diverse clinical isolates of *Staphylococcus aureus* LAC and UAMS-1, *Virulence* 11 (2020) 1738–1762, <https://doi.org/10.1080/21505594.2020.1855923>.
- [193] H.S. Lee, H.-S. Song, H.-J. Lee, S.H. Kim, M.J. Suh, J.Y. Cho, S. Ham, Y.-G. Kim, H.-S. Joo, W. Kim, S.H. Lee, D. Yoo, S.K. Bhatia, Y.-H. Yang, Comparative study of the difference in behavior of the accessory gene regulator (Agr) in USA300 and USA400 community-associated methicillin-resistant *Staphylococcus aureus* (CA-MRSA), *J. Microbiol. Biotechnol.* 31 (2021) 1060–1068, <https://doi.org/10.4014/jmb.2104.04032>.
- [194] S. Poudel, H. Tsunemoto, Y. Seif, A.V. Sastry, R. Szubin, S. Xu, H. Machado, C. A. Olson, A. Anand, J. Pogliano, V. Nizet, B.O. Palsson, Revealing 29 sets of independently modulated genes in *Staphylococcus aureus*, their regulators, and role in key physiological response, *Proc. Natl. Acad. Sci.* 117 (2020) 17228–17239, <https://doi.org/10.1073/pnas.2008413117>.
- [195] S.E. Rowe, N.J. Wagner, L. Li, J.E. Beam, A.D. Wilkinson, L.C. Radlinski, Q. Zhang, E.A. Miao, B.P. Conlon, Reactive oxygen species induce antibiotic tolerance during systemic *Staphylococcus aureus* infection, *Nat. Microbiol.* 5 (2019) 282–290, <https://doi.org/10.1038/s41564-019-0627-y>.
- [196] M. Mohammad, A. Ali, M.-T. Nguyen, F. Götz, R. Pullerits, T. Jin, *Staphylococcus aureus* lipoproteins in infectious diseases, *Front. Microbiol.* 13 (2022), <https://doi.org/10.3389/fmicb.2022.1006765>.
- [197] P.R. Hall, B.O. Elmore, C.H. Spang, S.M. Alexander, B.C. Manifold-Wheeler, M. J. Castleman, S.M. Daly, M.M. Peterson, E.K. Sully, J.K. Fleming, M. Otto, A. R. Horswill, G.S. Timmins, H.D. Gresham, Nox2 modification of LDL is essential for optimal apolipoprotein B-mediated control of agr type III *Staphylococcus aureus* quorum-sensing, *PLoS Pathog.* 9 (2013) e1003166.
- [198] S.E. George, J. Hrubesch, I. Breuing, N. Vetter, N. Korn, K. Hennemann, L. Bleul, M. Willmann, P. Ebner, F. Götz, C. Wolz, Oxidative stress drives the selection of quorum sensing mutants in the *Staphylococcus aureus* population, *Proc. Natl. Acad. Sci.* 116 (2019) 19145–19154, <https://doi.org/10.1073/pnas.1902752116>.
- [199] J. Baković, B.Y.K. Yu, D. Silva, M. Baczyńska, S.Y. Peak-Chew, A. Switzer, L. Burchell, S. Wigneshweraraj, M. Vandanasree, B. Gopal, V. Filonenko, M. Skehel, I. Gout, Redox regulation of the quorum-sensing transcription factor AgrA by coenzyme A, *Antioxidants* 10 (2021) 841, <https://doi.org/10.3390/antiox10060841>.
- [200] N. Linzner, V. Van Loi, V.N. Fritsch, H. Antelmam, Thiol-based redox switches in the major pathogen *Staphylococcus aureus*, *Biol. Chem.* 402 (2021) 333–361, <https://doi.org/10.1515/hsz-2020-0272>.
- [201] F. Sun, Y. Ding, Q. Ji, Z. Liang, X. Deng, C.C.L. Wong, C. Yi, L. Zhang, S. Xie, S. Alvarez, L.M. Hicks, C. Luo, H. Jiang, L. Lan, C. He, Protein cysteine phosphorylation of SarA/MgrA family transcriptional regulators mediates bacterial virulence and antibiotic resistance, *Proc. Natl. Acad. Sci.* 109 (2012) 15461–15466, <https://doi.org/10.1073/pnas.1205952109>.
- [202] R. Ma, X. Hu, X. Zhang, W. Wang, J. Sun, Z. Su, C. Zhu, Strategies to prevent, curb and eliminate biofilm formation based on the characteristics of various periods in one biofilm life cycle, *Front. Cell. Infect. Microbiol.* 12 (2022), <https://doi.org/10.3389/fcimb.2022.1003033>.
- [203] R.K. Gupta, T.T. Luong, C.Y. Lee, RNAIII of the *Staphylococcus aureus* agr system activates global regulator MgrA by stabilizing mRNA, *Proc. Natl. Acad. Sci.* 112 (2015) 14036–14041, <https://doi.org/10.1073/pnas.1509251112>.
- [204] E.Y. Bassy, M. Walch, D. Martinvalet, Reactive oxygen species: do they play a role in adaptive immunity? *Front. Immunol.* 12 (2021), <https://doi.org/10.3389/fimmu.2021.755856>.
- [205] M. Aubourg, M. Pottier, A. Léon, B. Bernay, A. Dhalluin, M. Cacaci, R. Torelli, P. Ledormand, C. Martini, M. Sanguinetti, M. Auzou, F. Gravey, J.-C. Giard, Inactivation of the response regulator AgrA Has a pleiotropic effect on biofilm formation, pathogenesis and stress response in *Staphylococcus lugdunensis*, *Microbiol. Spectr.* 10 (2022), <https://doi.org/10.1128/spectrum.01598-21>.
- [206] S. Amin Yavari, S.M. Castenmiller, J.A.G. van Strijp, M. Croes, Combating implant infections: shifting focus from bacteria to host, *Adv. Mater.* 32 (2020), <https://doi.org/10.1002/adma.202002962>.
- [207] E.H. James, A.M. Edwards, S. Wigneshweraraj, Transcriptional downregulation of agr expression in *Staphylococcus aureus* during growth in human serum can be overcome by constitutively active mutant forms of the sensor kinase AgrC, *FEMS Microbiol. Lett.* 349 (2013) 153–162, <https://doi.org/10.1111/1574-6968.12309>.
- [208] M. Matsumoto, S. Nakagawa, L. Zhang, Y. Nakamura, A.E. Villaruz, M. Otto, C. Wolz, N. Inohara, G. Núñez, Interaction between *Staphylococcus* Agr virulence and neutrophils regulates pathogen expansion in the skin, *Cell Host Microbe* 29 (2021) 930–940.e4, <https://doi.org/10.1016/j.chom.2021.03.007>.
- [209] E.G. Vozza, M.E. Mulcahy, R.M. McLoughlin, Making the most of the host; targeting the autophagy pathway facilitates *Staphylococcus aureus* intracellular survival in neutrophils, *Front. Immunol.* 12 (2021), <https://doi.org/10.3389/fimmu.2021.667387>.
- [210] N.G. Naga, D.E. El-Badan, K.M. Ghanem, M.I. Shaaban, It is the time for quorum sensing inhibition as alternative strategy of antimicrobial therapy, *Cell Commun. Signal* 21 (2023) 133, <https://doi.org/10.1186/s12964-023-01154-9>.
- [211] B.O. Elmore, K.D. Triplett, P.R. Hall, Apolipoprotein B48, the structural component of chylomicrons is sufficient to antagonize *Staphylococcus aureus* quorum-sensing, *PLoS One* 10 (2015) e0125027, <https://doi.org/10.1371/journal.pone.0125027>.
- [212] J. Canovas, M. Baldry, M.S. Bojer, P.S. Andersen, B.H. Gless, P.K. Grzeskowiak, M. Stegger, P. Damborg, C.A. Olsen, H. Ingmer, Cross-talk between *Staphylococcus aureus* and other staphylococcal species via the agr quorum sensing system, *Front. Microbiol.* 7 (2016), <https://doi.org/10.3389/fmicb.2016.01733>.
- [213] V. Delgado-Rizo, M.A. Martínez-Guzmán, L. Iñiguez-Gutiérrez, A. García-Orozco, A. Alvarado-Navarro, M. Fafutis-Morris, Neutrophil extracellular traps and its

- implications in inflammation: an overview, *Front. Immunol.* 8 (2017), <https://doi.org/10.3389/fimmu.2017.00081>.
- [214] E. Schicke, Z. Cseresnyés, K. Rennert, V. Vau, K.F. Haupt, F. Hornung, S. Nietzsche, F. Swiczak, M. Schmidtko, B. Glück, M. Koch, M. Schacke, R. Heller, A.S. Mosig, M.T. Figge, C. Ehrhardt, B. Löffler, S. Deinhardt-Emmer, *Staphylococcus aureus* lung infection results in down-regulation of surfactant protein-a mainly caused by pro-inflammatory macrophages, *Microorganisms* 8 (2020) 577, <https://doi.org/10.3390/microorganisms8040577>.
- [215] P. Magdalena, P.A. I.P. Gregory, P. Andrew, K. Jisun, D. Ashley, Z. Erin, U.R.J.K. T. K.Z. Chunyi, H.A.F.S. Julia, W.G.A.K. Junbeom, C. John, R.A. R.W.J.N.N.C. R.L. D. S.P. Dane, P. Alejandro, Z. Xilin, D. Karl, Y. Itai, T.V. J., S. Bo, Quorum-sensing agr system of *Staphylococcus aureus* primes gene expression for protection from lethal oxidative stress, *Elife*. 12 (2023) 1–52, <https://elifesciences.org/reviewed-preprints/89098>.
- [216] N. Tiwari, M. López-Redondo, L. Miguel-Romero, K. Kulhankova, M.P. Cahill, P. M. Tran, K.J. Kinney, S.H. Kilgore, H. Al-Tameemi, C.A. Herfst, S.W. Tuffs, J. R. Kirby, J.M. Boyd, J.K. McCormick, W. Salgado-Pabón, A. Marina, P. M. Schlievert, E.J. Fuentes, The SrrAB two-component system regulates *Staphylococcus aureus* pathogenicity through redox sensitive cysteines, *Proc. Natl. Acad. Sci.* 117 (2020) 10989–10999, <https://doi.org/10.1073/pnas.1921307117>.
- [217] G. Efthimiou, G. Tsiamis, M.A. Typas, K.M. Pappas, Transcriptomic adjustments of *Staphylococcus aureus* COL (MRSA) forming biofilms under acidic and alkaline conditions, *Front. Microbiol.* 10 (2019), <https://doi.org/10.3389/fmicb.2019.02393>.
- [218] M. Cordero, J. García-Fernández, I.C. Acosta, A. Yepes, J. Avendano-Ortiz, C. Lisowski, B. Oesterreich, K. Ohlsen, E. Lopez-Collazo, K.U. Förstner, A. Eulalio, D. Lopez, The induction of natural competence adapts staphylococcal metabolism to infection, *Nat. Commun.* 13 (2022) 1525, <https://doi.org/10.1038/s41467-022-29206-7>.
- [219] X. Shu, Y. Shi, Y. Huang, D. Yu, B. Sun, Transcription tuned by S-nitrosylation underlies a mechanism for *Staphylococcus aureus* to circumvent vancomycin killing, *Nat. Commun.* 14 (2023) 2318, <https://doi.org/10.1038/s41467-023-37949-0>.
- [220] F. Chen, H. Di, Y. Wang, C. Peng, R. Chen, H. Pan, C.-G. Yang, H. Liang, L. Lan, The enzyme activity of sortase A is regulated by phosphorylation in *Staphylococcus aureus*, *Virulence* 14 (2023), <https://doi.org/10.1080/21505594.2023.2171641>.
- [221] D. Balasubramanian, L. Harper, B. Shopsis, V.J. Torres, *Staphylococcus aureus* pathogenesis in diverse host environments, *Pathog. Dis.* (2017) ftx005, <https://doi.org/10.1093/femspd/ftx005>.
- [222] N.R. Waters, D.J. Samuels, R.K. Behera, J. Livny, K.Y. Rhee, M.R. Sadykov, S. R. Brinsmade, A spectrum of CodY activities drives metabolic reorganization and virulence gene expression in *Staphylococcus aureus*, *Mol. Microbiol.* 101 (2016) 495–514, <https://doi.org/10.1111/mmi.13404>.
- [223] J.C. Kaiser, A.N. King, J.C. Grigg, J.R. Sheldon, D.R. Edgell, M.E.P. Murphy, S. R. Brinsmade, D.E. Heinrichs, Repression of branched-chain amino acid synthesis in *Staphylococcus aureus* is mediated by isoleucine via CodY, and by a leucine-rich attenuator peptide, *PLoS Genet.* 14 (2018) e1007159.
- [224] A. Han, H.-R. Kang, J. Son, D.H. Kwon, S. Kim, W.C. Lee, H.K. Song, M.J. Song, K. Y. Hwang, The structure of the pleiotropic transcription regulator CodY provides insight into its GTP-sensing mechanism, *Nucleic Acids Res.* (2016) gkw775, <https://doi.org/10.1093/nar/gkw775>.
- [225] K. Schilcher, A.R. Horswill, *Staphylococcal* biofilm development: structure, regulation, and treatment strategies, *Microbiol. Mol. Biol. Rev.* 84 (2020), <https://doi.org/10.1128/MMBR.00026-19>.
- [226] A.N. King, S.A. Borkar, D.J. Samuels, S. Batz, L.L. Bullock, M.R. Sadykov, K. W. Bayles, S.R. Brinsmade, Guanine limitation results in CodY-Dependent and -independent alteration of *Staphylococcus aureus* physiology and gene expression, *J. Bacteriol.* 200 (2018), <https://doi.org/10.1128/JB.00136-18>.
- [227] P. Dauros-Singorenko, S. Wiles, S. Swift, *Staphylococcus aureus* biofilms and their response to a relevant in vivo iron source, *Front. Microbiol.* 11 (2020), <https://doi.org/10.3389/fmicb.2020.509525>.
- [228] A.S. Vasilchenko, E.A. Rogozhin, Sub-inhibitory effects of antimicrobial peptides, *Front. Microbiol.* 10 (2019), <https://doi.org/10.3389/fmicb.2019.01160>.
- [229] D. Bronesky, E. Desgranges, A. Corvaglia, P. François, C.J. Caballero, L. Prado, A. Toledo-Arana, I. Lasa, K. Moreau, F. Vandenesch, S. Marzi, P. Romby, I. Caldelari, A multifaceted small <sc>scRNA</sc> modulates gene expression upon glucose limitation in *Staphylococcus aureus*, *EMBO J.* 38 (2019), <https://doi.org/10.15252/embj.201899363>.
- [230] Z. Luo, S. Yue, T. Chen, P. She, Y. Wu, Y. Wu, Reduced growth of *Staphylococcus aureus* under high glucose conditions is associated with decreased pentaglycine expression, *Front. Microbiol.* 11 (2020), <https://doi.org/10.3389/fmicb.2020.537290>.
- [231] Z.R. Tranchemontagne, R.B. Camire, V.J. O'Donnell, J. Baugh, K.M. Burkholder, *Staphylococcus aureus* strain USA300 perturbs acquisition of lysosomal enzymes and requires phagosomal acidification for survival inside macrophages, *Infect. Immun.* 84 (2016) 241–253, <https://doi.org/10.1128/IAI.00704-15>.
- [232] M.R. Hauserman, M.J. Ferraro, R.K. Carroll, K.C. Rice, Altered quorum sensing and physiology of *Staphylococcus aureus* during spaceflight detected by multi-omics data analysis, *Npj Microgravity* 10 (2024) 2, <https://doi.org/10.1038/s41526-023-00343-7>.
- [233] R.A. Bastock, E.C. Marino, R.E. Wiemels, D.L. Holzschu, R.A. Keogh, R.L. Zapf, E. R. Murphy, R.K. Carroll, *Staphylococcus aureus* responds to physiologically relevant temperature changes by altering its global transcript and protein profile, *MSphere* 6 (2021), <https://doi.org/10.1128/mSphere.01303-20>.
- [234] P. Fajardo-Cavazos, W.L. Nicholson, Establishing standard protocols for bacterial culture in biological research in canisters (BRIC) hardware, *Gravitational Sp. Res.* 4 (2016) 58–69, <https://doi.org/10.2478/gsr-2016-0013>.
- [235] H. Hu, D. Zhong, W. Li, X. Lin, J. He, Y. Sun, Y. Wu, M. Shi, X. Chen, F. Xu, M. Zhou, Microalgae-based bioactive hydrogel loaded with quorum sensing inhibitor promotes infected wound healing, *Nano Today* 42 (2022) 101368, <https://doi.org/10.1016/j.nantod.2021.101368>.
- [236] K. Vadakkan, S. Vijayanand, J. Hemapriya, R. Gunasekaran, Quorum quenching attenuation of rhizobacteria *Chromobacterium violaceum* virulence by *Desmodium gangeticum* root extract, *Rhizosphere*. 22 (2022) 100490, <https://doi.org/10.1016/j.rhisph.2022.100490>.
- [237] M. Ciulla, A. Di Stefano, L. Marinelli, I. Cacciatori, G. Di Biase, RNAIII inhibiting peptide (RIP) and derivatives as potential tools for the treatment of *S. aureus* biofilm infections, *Curr. Top. Med. Chem.* 18 (2019) 2068–2079, <https://doi.org/10.2174/1568026618666181022120711>.
- [238] J. Dong, W. Wang, W. Zhou, S. Zhang, M. Li, N. Li, G. Pan, X. Zhang, J. Bai, C. Zhu, Immunomodulatory biomaterials for implant-associated infections: from conventional to advanced therapeutic strategies, *Biomater. Res.* 26 (2022) 72, <https://doi.org/10.1186/s40824-022-00326-x>.
- [239] M. Tamai, Y. Yamazaki, T. Ito, S. Nakagawa, Y. Nakamura, Pathogenic role of the staphylococcal accessory gene regulator quorum sensing system in atopic dermatitis, *Front. Cell. Infect. Microbiol.* 13 (2023), <https://doi.org/10.3389/fcimb.2023.1178650>.
- [240] S. Sugathan, N.S. Pradeep, S. Abdulhameed, eds., *Bioresources and Bioprocess in Biotechnology*, Springer Singapore, Singapore, 2017, <https://doi.org/10.1007/978-981-10-4284-3>.
- [241] Y. Igarashi, F. Gohda, T. Kadoshima, T. Fukuda, T. Hanafusa, A. Shojima, J. Nakayama, G.F. Bills, S. Peterson, Avellanin C, an inhibitor of quorum-sensing signaling in *Staphylococcus aureus*, from *Hamigera ingelheimensis*, *J. Antibiot.* (Tokyo) 68 (2015) 707–710, <https://doi.org/10.1038/ja.2015.50>.
- [242] D.A. Todd, C.P. Parlet, H.A. Crosby, C.L. Malone, K.P. Heilmann, A.R. Horswill, N. B. Cech, Signal biosynthesis inhibition with ambuic acid as a strategy to target antibiotic-resistant infections, *Antimicrob. Agents Chemother.* 61 (2017), <https://doi.org/10.1128/AAC.00263-17>.
- [243] F. Tang, L. Li, X.-M. Meng, B. Li, C.-Q. Wang, S.-Q. Wang, T.-L. Wang, Y.-M. Tian, Inhibition of alpha-hemolysin expression by resveratrol attenuates *Staphylococcus aureus* virulence, *Microb. Pathog.* 127 (2019) 85–90, <https://doi.org/10.1016/j.micpath.2018.11.027>.
- [244] A. Muhs, J.T. Lyles, C.P. Parlet, K. Nelson, J.S. Kavanaugh, A.R. Horswill, C. L. Quave, Virulence inhibitors from brazilian peppertree block quorum sensing and abate dermonecrosis in skin infection models, *Sci. Rep.* 7 (2017) 42275, <https://doi.org/10.1038/srep42275>.
- [245] B. Palaniappan, A.P. Solomon, Targeting AgrA quorum sensing regulator by bumetanide attenuates virulence in *Staphylococcus aureus* – a drug repurposing approach, *Life Sci.* 273 (2021) 119306, <https://doi.org/10.1016/j.lfs.2021.119306>.
- [246] N.H. Mahdally, R.F. George, M.T. Kashef, M. Al-Ghobashy, F.E. Murad, A.S. Attia, Staquorsin: a novel *Staphylococcus aureus* Agr-mediated quorum sensing inhibitor impairing virulence in vivo without notable resistance development, *Front. Microbiol.* 12 (2021), <https://doi.org/10.3389/fmicb.2021.700494>.
- [247] X. Ren, X. Guo, C. Liu, S. Jing, T. Wang, L. Wang, J. Guan, W. Song, Y. Zhao, Y. Shi, Natural flavone hispidulin protects mice from *Staphylococcus aureus* pneumonia by inhibition of α -hemolysin production via targeting AgrAC, *Microbiol. Res.* 261 (2022) 127071, <https://doi.org/10.1016/j.micres.2022.127071>.
- [248] V. Khodaverdian, M. Pesho, B. Truitt, L. Bollinger, P. Patel, S. Nithianantham, G. Yu, E. Delaney, E. Jankowsky, M. Shoham, Discovery of antivirulence agents against methicillin-resistant *Staphylococcus aureus*, *Antimicrob. Agents Chemother.* 57 (2013) 3645–3652, <https://doi.org/10.1128/AAC.00269-13>.
- [249] Q. Yuan, W. Feng, Y. Wang, Q. Wang, N. Mou, L. Xiong, X. Wang, P. Xia, F. Sun, Luteolin attenuates the pathogenesis of *Staphylococcus aureus* by interfering with the agr system, *Microb. Pathog.* 165 (2022) 105496, <https://doi.org/10.1016/j.micpath.2022.105496>.
- [250] G. Brackman, K. Breyne, R. De Rycke, A. Vermote, F. Van Nieuwerburgh, E. Meyer, S. Van Calenbergh, T. Coenye, The quorum sensing inhibitor hamamelitannin increases antibiotic susceptibility of *Staphylococcus aureus* biofilms by affecting peptidoglycan biosynthesis and eDNA release, *Sci. Rep.* 6 (2016) 20321, <https://doi.org/10.1038/srep20321>.
- [251] E.K. Sully, N. Malachowa, B.O. Elmore, S.M. Alexander, J.K. Fleming, B.M. Gray, F.R. DeLeo, M. Otto, A.L. Cheung, B.S. Edwards, L.A. Sklar, A.R. Horswill, P. R. Hall, H.D. Gresham, Selective chemical inhibition of agr quorum sensing in *Staphylococcus aureus* promotes host defense with minimal impact on resistance, *PLoS Pathog.* 10 (2014) e1004174.
- [252] A.M. Salam, C.L. Quave, Targeting virulence in *Staphylococcus aureus* by chemical inhibition of the accessory gene regulator system in vivo, *MSphere* 3 (2018), <https://doi.org/10.1128/mSphere.00500-17>.
- [253] S.M. Daly, B.O. Elmore, J.S. Kavanaugh, K.D. Triplett, M. Figueroa, H.A. Raja, T. El-Elmait, H.A. Crosby, J.K. Fleming, N.B. Cech, A.R. Horswill, N.H. Oberlies, P.R. Hall, ω -hydroxyemodin limits *Staphylococcus aureus* quorum sensing-mediated pathogenesis and inflammation, *Antimicrob. Agents Chemother.* 59 (2015) 2223–2235, <https://doi.org/10.1128/AAC.04564-14>.
- [254] M. Baldry, Y. Nakamura, S. Nakagawa, D. Frees, H. Matsue, G. Núñez, H. Ingmer, Application of an agr-specific antivirulence compound as therapy for *Staphylococcus aureus*-induced inflammatory skin disease, *J. Infect. Dis.* 218 (2018) 1009–1013, <https://doi.org/10.1093/infdis/jiy259>.

- [255] N. Cech, H. Junio, L. Ackermann, J. Kavanaugh, A. Horswill, Quorum quenching and antimicrobial activity of goldenseal (*hydrastis canadensis*) against methicillin-resistant *Staphylococcus aureus* (MRSA), *Planta Med.* 78 (2012) 1556–1561, <https://doi.org/10.1055/s-0032-1315042>.
- [256] Y. Zhao, J. Wei, C. Li, A.F. Ahmed, Z. Liu, C. Ma, A comprehensive review on mechanism of natural products against *Staphylococcus aureus*, *J. Futur. Foods* 2 (2022) 25–33, <https://doi.org/10.1016/j.jfutfo.2022.03.014>.
- [257] C. Kong, H. Neoh, S. Nathan, Targeting *Staphylococcus aureus* toxins: a potential form of anti-virulence therapy, *Toxins (Basel)* 8 (2016) 72, <https://doi.org/10.3390/toxins8030072>.
- [258] M.-K. Kim, *Staphylococcus aureus* toxins: from their pathogenic roles to anti-virulence therapy using natural products, *Biotechnol. Bioprocess Eng.* 24 (2019) 424–435, <https://doi.org/10.1007/s12257-019-0059-9>.
- [259] J. Wang, X. Zhou, S. Liu, G. Li, L. Shi, J. Dong, W. Li, X. Deng, X. Niu, Morin hydrate attenuates *Staphylococcus aureus* virulence by inhibiting the self-assembly of α -hemolysin, *J. Appl. Microbiol.* 118 (2015) 753–763, <https://doi.org/10.1111/jam.12743>.
- [260] P. Patel, K. Parmar, M. Das, Inhibition of insulin amyloid fibrillation by Morin hydrate, *Int. J. Biol. Macromol.* 108 (2018) 225–239, <https://doi.org/10.1016/j.ijbiomac.2017.11.168>.
- [261] J. Qiu, D. Wang, Y. Zhang, J. Dong, J. Wang, X. Niu, Molecular modeling reveals the novel inhibition mechanism and binding mode of three natural compounds to staphylococcal α -hemolysin, *PLoS One* 8 (2013) e80197, <https://doi.org/10.1371/journal.pone.0080197>.
- [262] J. Dong, J. Qiu, Y. Zhang, C. Lu, X. Dai, J. Wang, H. Li, X. Wang, W. Tan, M. Luo, X. Niu, X. Deng, Oroxylin A inhibits hemolysis via hindering the self-assembly of α -hemolysin heptameric transmembrane pore, *PLoS Comput. Biol.* 9 (2013) e1002869, <https://doi.org/10.1371/journal.pcbi.1002869>.
- [263] L. Jiang, H. Li, L. Wang, Z. Song, L. Shi, W. Li, X. Deng, J. Wang, Isorhamnetin attenuates *Staphylococcus aureus*-induced lung cell injury by inhibiting alpha-hemolysin expression, *J. Microbiol. Biotechnol.* 26 (2016) 596–602, <https://doi.org/10.4014/jmb.1507.07091>.
- [264] S.F. Nabavi, N. Braid, S. Habtemariam, I.E. Orhan, M. Daglia, A. Manayi, O. Gortzi, S.M. Nabavi, Neuroprotective effects of chrysin: from chemistry to medicine, *Neurochem. Int.* 90 (2015) 224–231, <https://doi.org/10.1016/j.neuint.2015.09.006>.
- [265] Y. Zhou, H. Zhang, C. Peng, Puerarin: a review of pharmacological effects, *Phyther. Res.* 28 (2014) 961–975, <https://doi.org/10.1002/ptr.5083>.
- [266] F. Tang, W.-H. Li, X. Zhou, Y.-H. Liu, Z. Li, Y.-S. Tang, X. Kou, S.-D. Wang, M. Bao, L.-D. Qu, M. Li, B. Li, Puerarin protects against *Staphylococcus aureus*-induced injury of human alveolar epithelial A549 cells via downregulating alpha-hemolysin secretion, *Microb. Drug Resist.* 20 (2014) 357–363, <https://doi.org/10.1089/mdr.2013.0104>.
- [267] A.W. Khan, S. Kotta, S.H. Ansari, R.K. Sharma, J. Ali, Self-nanoemulsifying drug delivery system (SNEDDS) of the poorly water-soluble grapefruit flavonoid Naringenin: design, characterization, in vitro and in vivo evaluation, *Drug Deliv.* 22 (2015) 552–561, <https://doi.org/10.3109/10717544.2013.878003>.
- [268] B. Salehi, P. Fokou, M. Sharifi-Rad, P. Zucca, R. Pezzani, N. Martins, J. Sharifi-Rad, The therapeutic potential of naringenin: a review of clinical trials, *Pharmaceuticals* 12 (2019) 11, <https://doi.org/10.3390/ph12010011>.
- [269] Y. Zhang, J. Wang, J. Dong, J. Wei, Y. Wang, X. Dai, X. Wang, M. Luo, W. Tan, X. Deng, X. Niu, Inhibition of α -toxin production by subinhibitory concentrations of naringenin controls *Staphylococcus aureus* pneumonia, *Fitoaterapia* 86 (2013) 92–99, <https://doi.org/10.1016/j.fito.2013.02.001>.
- [270] M. Figueroa, A.K. Jarmusch, H.A. Raja, T. El-Elm, J.S. Kavanaugh, A. R. Horswill, R.G. Cooks, N.B. Cech, N.H. Oberlies, Polyhydroxyanthraquinones as quorum sensing inhibitors from the guttates of penicillium restrictum and their analysis by desorption electrospray ionization mass spectrometry, *J. Nat. Prod.* 77 (2014) 1351–1358, <https://doi.org/10.1021/np5000704>.
- [271] C.L. Quave, J.T. Lyles, J.S. Kavanaugh, K. Nelson, C.P. Parlet, H.A. Crosby, K. P. Heilmann, A.R. Horswill, Castanea sativa (European Chestnut) leaf extracts rich in ursene and oleanene derivatives block staphylococcus aureus virulence and pathogenesis without detectable resistance, *PLoS One* 10 (2015) e0136486, <https://doi.org/10.1371/journal.pone.0136486>.
- [272] J. Li, Q. Fan, M. Jin, C. Mao, H. Zhang, X. Zhang, L. Sun, D. Grenier, L. Yi, X. Hou, Y. Wang, Paeoniflorin reduce luxS/AI-2 system-controlled biofilm formation and virulence in *Streptococcus suis*, *Virulence* 12 (2021) 3062–3073, <https://doi.org/10.1080/21505594.2021.2010398>.
- [273] H. Zhang, Z. Zhang, J. Li, G. Qin, New Strategies for Biocontrol of Bacterial Toxins and Virulence: Focusing on Quorum-Sensing Interference and Biofilm Inhibition, *Toxins (Basel)* 15 (2023) 570, <https://doi.org/10.3390/toxins15090570>.
- [274] T. Vij, P.P. Anil, R. Shams, K.K. Dash, R. Kalsi, V.K. Pandey, E. Harsányi, B. Kovács, A.M. Shaikh, A comprehensive review on bioactive compounds found in caesalpinia sappan, *Molecules* 28 (2023) 6247, <https://doi.org/10.3390/molecules28176247>.
- [275] B. Wang, P.-W. Wei, S. Wan, Y. Yao, C.-R. Song, P.-P. Song, G.-B. Xu, Z.-Q. Hu, Z. Zeng, C. Wang, H.-M. Liu, Ginkgo biloba exocarp extracts inhibit *S. aureus* and MRSA by disrupting biofilms and affecting gene expression, *J. Ethnopharmacol.* 271 (2021) 113895, <https://doi.org/10.1016/j.jep.2021.113895>.
- [276] T. Lin, L. Huang, N. Cheng, Y. Wang, Z. Ning, S. Huang, Y. Wu, T. Chen, S. Su, Y. Lin, The in vitro and in vivo antibacterial activities of uniflorous honey from a medicinal plant, *Scrophularia ningpoensis* Hemsl., and characterization of its chemical profile with UPLC-MS/MS, *J. Ethnopharmacol.* 296 (2022) 115499, <https://doi.org/10.1016/j.jep.2022.115499>.
- [277] X. Bai, Y. Shen, T. Zhang, R. Meng, Y. Zhang, Y. Deng, N. Guo, Anti-biofilm activity of biochanin A against *Staphylococcus aureus*, *Appl. Microbiol. Biotechnol.* 107 (2023) 867–879, <https://doi.org/10.1007/s00253-022-12350-x>.
- [278] X. Wu, G.-L. Ma, H.-W. Chen, Z.-Y. Zhao, Z.-P. Zhu, J. Xiong, G.-X. Yang, J.-F. Hu, Antibacterial and antibiofilm efficacy of the preferred fractions and compounds from *Euphorbia humifusa* (herba euphorbiae humifusae) against *Staphylococcus aureus*, *J. Ethnopharmacol.* 306 (2023) 116177, <https://doi.org/10.1016/j.jep.2023.116177>.
- [279] D. Ming, D. Wang, F. Cao, H. Xiang, D. Mu, J. Cao, B. Li, L. Zhong, X. Dong, X. Zhong, L. Wang, T. Wang, Kaempferol inhibits the primary attachment phase of biofilm formation in *Staphylococcus aureus*, *Front. Microbiol.* 8 (2017), <https://doi.org/10.3389/fmicb.2017.02263>.
- [280] B. Zhang, Z. Teng, X. Li, G. Lu, X. Deng, X. Niu, J. Wang, Chalcone attenuates *Staphylococcus aureus* virulence by targeting sortase A and alpha-hemolysin, *Front. Microbiol.* 8 (2017), <https://doi.org/10.3389/fmicb.2017.01715>.
- [281] G. Nitulescu, D. Margina, A. Zanfirescu, O.T. Olaru, G.M. Nitulescu, Targeting bacterial sortases in search of anti-virulence therapies with low risk of resistance development, *Pharmaceuticals* 14 (2021) 415, <https://doi.org/10.3390/ph14050415>.
- [282] X. Yan, S. Gu, Y. Shi, X. Cui, S. Wen, J. Ge, The effect of emodin on *Staphylococcus aureus* strains in planktonic form and biofilm formation in vitro, *Arch. Microbiol.* 199 (2017) 1267–1275, <https://doi.org/10.1007/s00203-017-1396-8>.
- [283] S.-J. Yum, H.G. Jeong, S.M. Kim, Anti-biofilm effects of sinomenine against *Staphylococcus aureus*, *Food Sci. Biotechnol.* 32 (2023) 83–90, <https://doi.org/10.1007/s10068-022-01174-0>.
- [284] A. Sharifi, A. Mohammadzadeh, T.Z. Salehi, P. Mahmoodi, A. Nourian, Cummin cymunol L. essential oil: a promising antibacterial and antiviral agent against multidrug-resistant *Staphylococcus aureus*, *Front. Microbiol.* 12 (2021), <https://doi.org/10.3389/fmicb.2021.667833>.
- [285] X. Guo, N. Mei, Aloe vera : a review of toxicity and adverse clinical effects, *J. Environ. Sci. Heal. Part C* 34 (2016) 77–96, <https://doi.org/10.1080/10590501.2016.1166826>.
- [286] B. Kis, F.C. Ifrim, V. Buda, S. Avram, I.Z. Pavel, D. Antal, V. Paunescu, C. A. Dehelean, F. Ardelean, Z. Diaconeasa, C. Soica, C. Danciu, Cannabidiol—from plant to human body: a promising bioactive molecule with multi-target effects in cancer, *Int. J. Mol. Sci.* 20 (2019) 5905, <https://doi.org/10.3390/ijms20235905>.
- [287] C.A. Ford, I.M. Hurford, J.E. Cassat, Antivirulence strategies for the treatment of *Staphylococcus aureus* infections: a mini review, *Front. Microbiol.* 11 (2021), <https://doi.org/10.3389/fmicb.2020.632706>.
- [288] B.A. Diep, J.J. Hilliard, V.T.M. Le, C. Tkaczky, H.N. Le, V.G. Tran, R.L. Rao, E. C. Dip, E.P. Pereira-Franchi, P. Cha, S. Jacobson, R. Broome, L.I. Cheng, W. Weiss, L. Prokai, V. Nguyen, C.K. Stover, B.R. Sellman, Targeting alpha toxin to mitigate its lethal toxicity in ferret and rabbit models of *Staphylococcus aureus* necrotizing pneumonia, *Antimicrob. Agents Chemother.* 61 (2017), <https://doi.org/10.1128/AAC.02456-16>.
- [289] X.-Q. Yu, G.J. Robbie, Y. Wu, M.T. Esser, K. Jensen, H.I. Schwartz, T. Bellamy, M. Hernandez-Illas, H.S. Jafri, Safety, tolerability, and pharmacokinetics of MED14893, an investigational, extended-half-life, anti-*Staphylococcus aureus* alpha-toxin human monoclonal antibody, in healthy adults, *Antimicrob. Agents Chemother.* 61 (2017), <https://doi.org/10.1128/AAC.01020-16>.
- [290] H. Rouha, A. Badarau, Z.C. Visram, M.B. Battles, B. Prinz, Z. Magyarys, G. Nagy, I. Mirkina, L. Stulik, M. Zerb, M. Jägerhofer, B. Maierhofer, A. Teubenbacher, I. Dolezilova, K. Gross, S. Banerjee, G. Zauner, S. Malafa, J. Zmajkovic, S. Maier, R. Mabry, E. Krauland, K.D. Witttrup, T.U. Gerngross, E. Nagy, Five birds, one stone: Neutralization of α -hemolysin and 4 bi-component leukocidins of *Staphylococcus aureus* with a single human monoclonal antibody, *mAbs* 7 (2015) 243–254, <https://doi.org/10.4161/19420862.2014.985132>.
- [291] L. Stulik, H. Rouha, D. Labrousse, Z.C. Visram, A. Badarau, B. Maierhofer, K. Groß, S. Weber, M.D. Kramaric, J. Gjojnaric, G. Nagy, D. Croisier, E. Nagy, Preventing lung pathology and mortality in rabbit *Staphylococcus aureus* pneumonia models with cytotoxin-neutralizing monoclonal IgGs penetrating the epithelial lining fluid, *Sci. Rep.* 9 (2019) 5339, <https://doi.org/10.1038/s41598-019-41826-6>.
- [292] C.P. Parlet, J.S. Kavanaugh, H.A. Raja, T. El-Elm, D.A. Todd, C. J. Pearce, N.B. Cech, N.H. Oberlies, A.R. Horswill, Apicidin attenuates MRSA virulence through quorum-sensing inhibition and enhanced host defense, *Cell Rep.* 27 (2019) 187–198.e6, <https://doi.org/10.1016/j.celrep.2019.03.018>.
- [293] A. Mukherjee, S. Bose, A. Shao, S.K. Das, Nanotechnology based therapeutic approaches: an advanced strategy to target the biofilm of ESKAPE pathogens, *Mater. Adv.* 4 (2023) 2544–2572, <https://doi.org/10.1039/D2MA00846G>.
- [294] K. Vadakkan, N.P. Rujit, A.K. Ngangbam, S. Vijayanand, N.K. Nedumpillai, Novel advancements in the sustainable green synthesis approach of silver nanoparticles (AgNPs) for antibacterial therapeutic applications, *Coord. Chem. Rev.* 499 (2024) 215528, <https://doi.org/10.1016/j.ccr.2023.215528>.
- [295] R. Kashapov, A. Ibragimova, R. Pavlov, D. Gabdrakhmanov, N. Kashapova, E. Burlilova, L. Zakharova, O. Sinyashin, Nanocarriers for biomedicine: from lipid formulations to inorganic and hybrid nanoparticles, *Int. J. Mol. Sci.* 22 (2021) 7055, <https://doi.org/10.3390/ijms22137055>.
- [296] F.M. Mosallam, H.A. Abbas, G.H. Shaker, S.E. Goma, Alleviating the virulence of *Pseudomonas aeruginosa* and *Staphylococcus aureus* by ascorbic acid nanoemulsion, *Philos. Phenomenol. Res. Microbiol.* 174 (2023) 104084, <https://doi.org/10.1016/j.resmic.2023.104084>.
- [297] A. Negi, K.K. Kesari, Chitosan nanoparticle encapsulation of antibacterial essential oils, *Micromachines* 13 (2022) 1265, <https://doi.org/10.3390/mi13081265>.

- [298] P. Paramanantham, B. Siddhardha, S.S.B. Lal, A. Sharan, A.A. Alyousef, M.S. Al Dosary, M. Arshad, A. Syed, Antimicrobial photodynamic therapy on *Staphylococcus aureus* and *Escherichia coli* using malachite green encapsulated mesoporous silica nanoparticles: an in vitro study, *PeerJ* 7 (2019) e7454, <https://doi.org/10.7717/peerj.7454>.
- [299] V. Pérez-Laguna, I. García-Luque, S. Ballesta, A. Rezusta, Y. Gilaberte, Photodynamic therapy combined with antibiotics or antifungals against microorganisms that cause skin and soft tissue infections: a planktonic and biofilm approach to overcome resistances, *Pharmaceuticals* 14 (2021) 603, <https://doi.org/10.3390/ph14070603>.
- [300] Y. Li, S. Li, K. Yang, R. Guo, X. Zhu, Y. Shi, A. Huang, Antibiofilm mechanism of a novel milk-derived antimicrobial peptide against *Staphylococcus aureus* by downregulating agr quorum sensing system, *J. Appl. Microbiol.* 133 (2022) 2198–2209, <https://doi.org/10.1111/jam.15653>.